Pune District, 1990–2022:

a 30-metre satellite record of land change

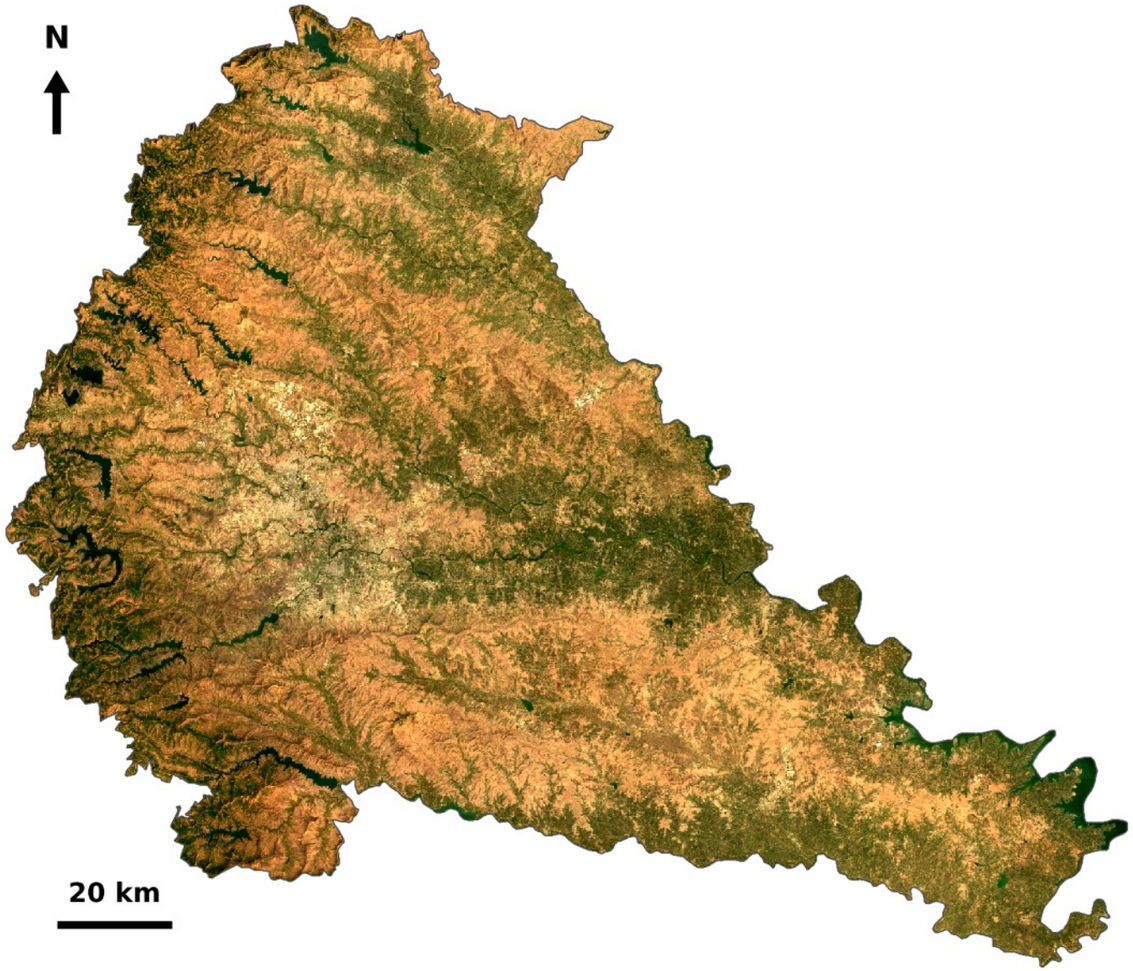
Aim: build one consistent 30 m record of land cover across the whole district from 1990 to 2022, and measure how far it can be trusted.

Arush · PhD visual tour of the Phase 7 district data · 9 October 2026

All pictures are drawn from the project's own files.

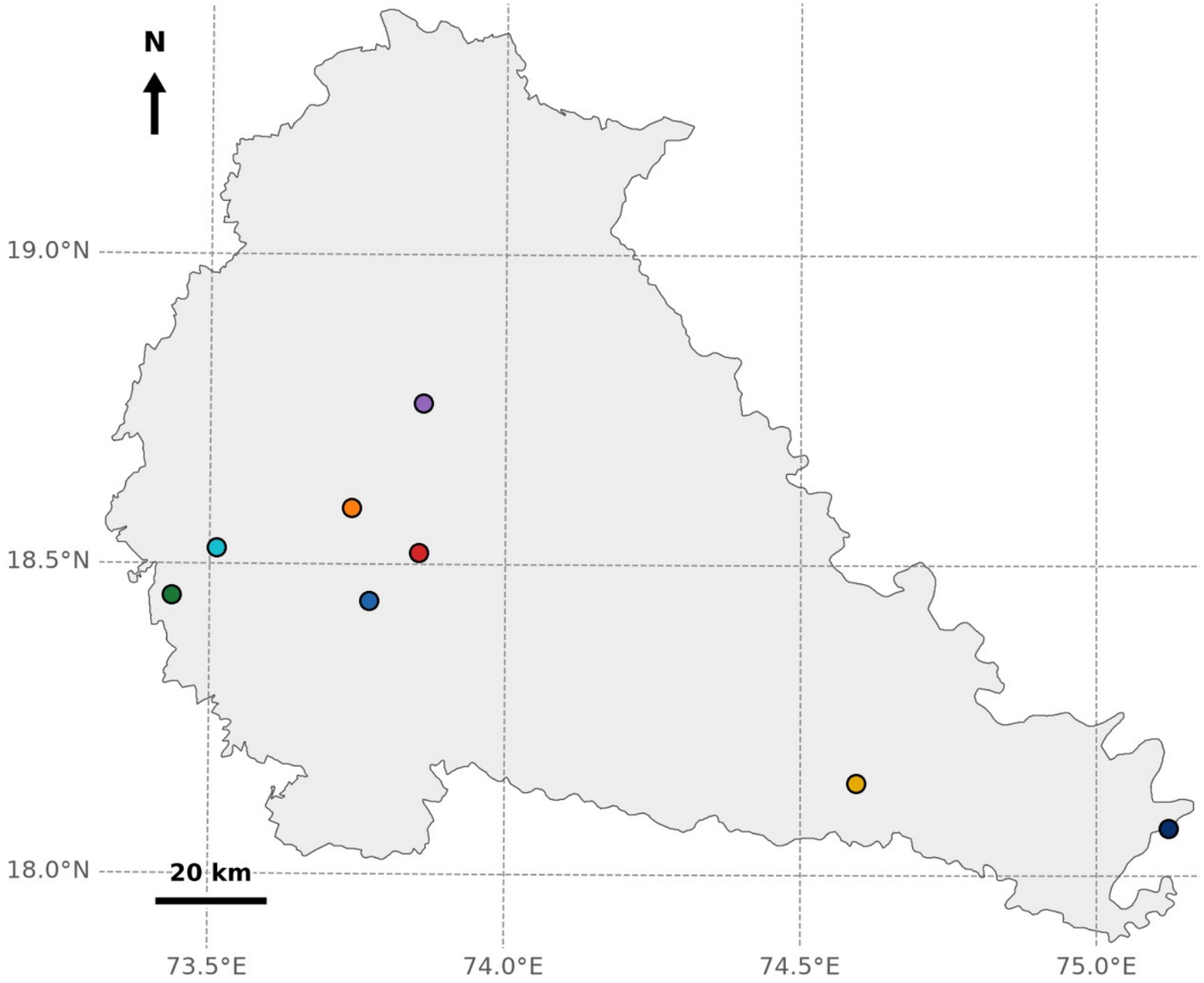
No accuracy numbers and no classification maps are shown.

Pune District, dry season 2020 (Landsat 7 + 8 true colour)



supervisor copy - do not share with interpreters

Where: Pune District on the 30 m grid

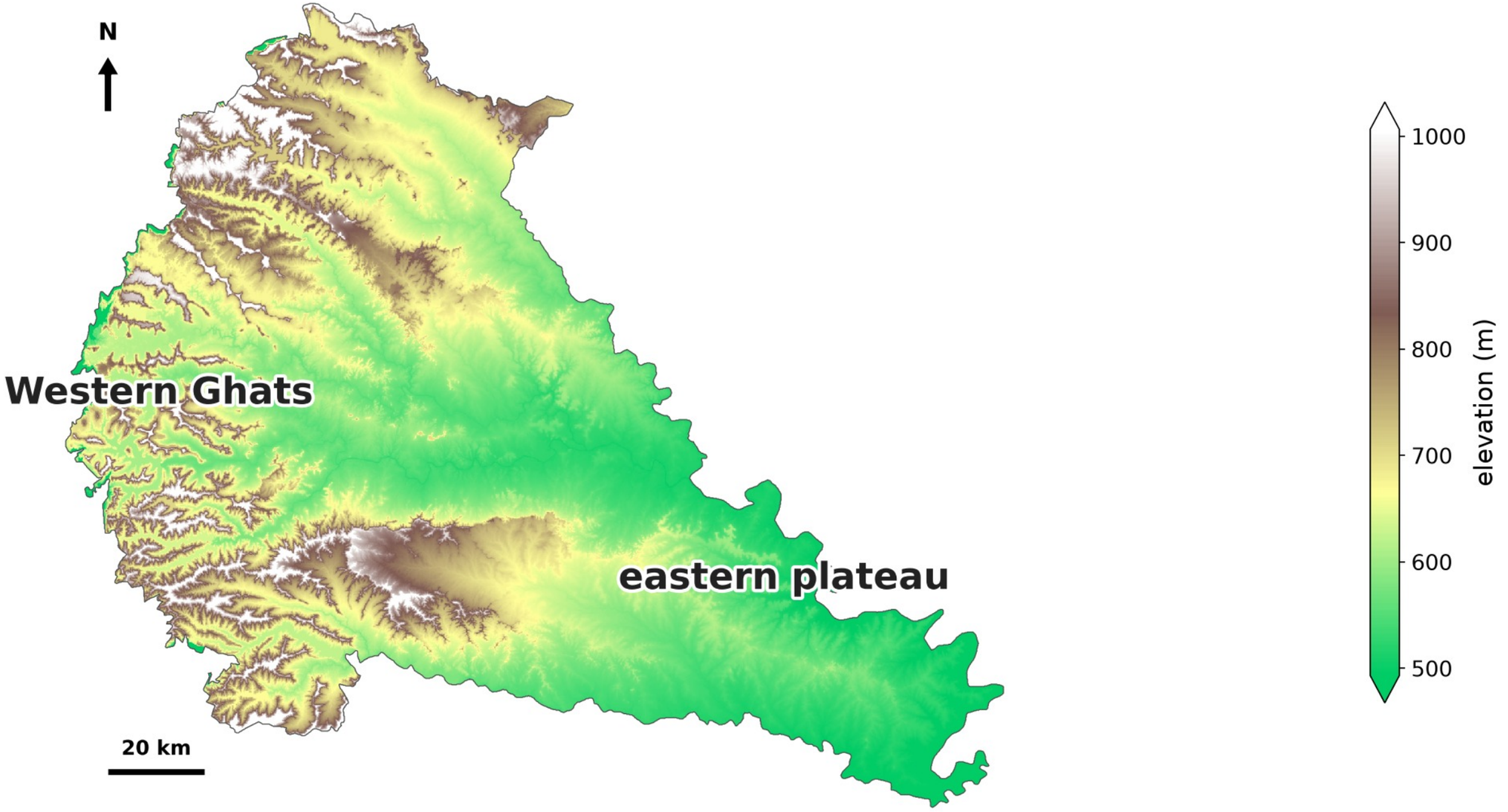


Places (final pixel centre)

- **Shaniwar Wada**
  18.5194° N, 73.8541° E
- **Khadakwasla**
  18.4403° N, 73.7695° E
- **Hinjewadi**
  18.5914° N, 73.7388° E
- **Chakan**
  18.7599° N, 73.8596° E
- **Baramati**
  18.1497° N, 74.5945° E
- **Mulshi**
  18.5252° N, 73.5111° E
- **Tamhini**
  18.4498° N, 73.4362° E
- **Ujani**
  18.0763° N, 75.1214° E

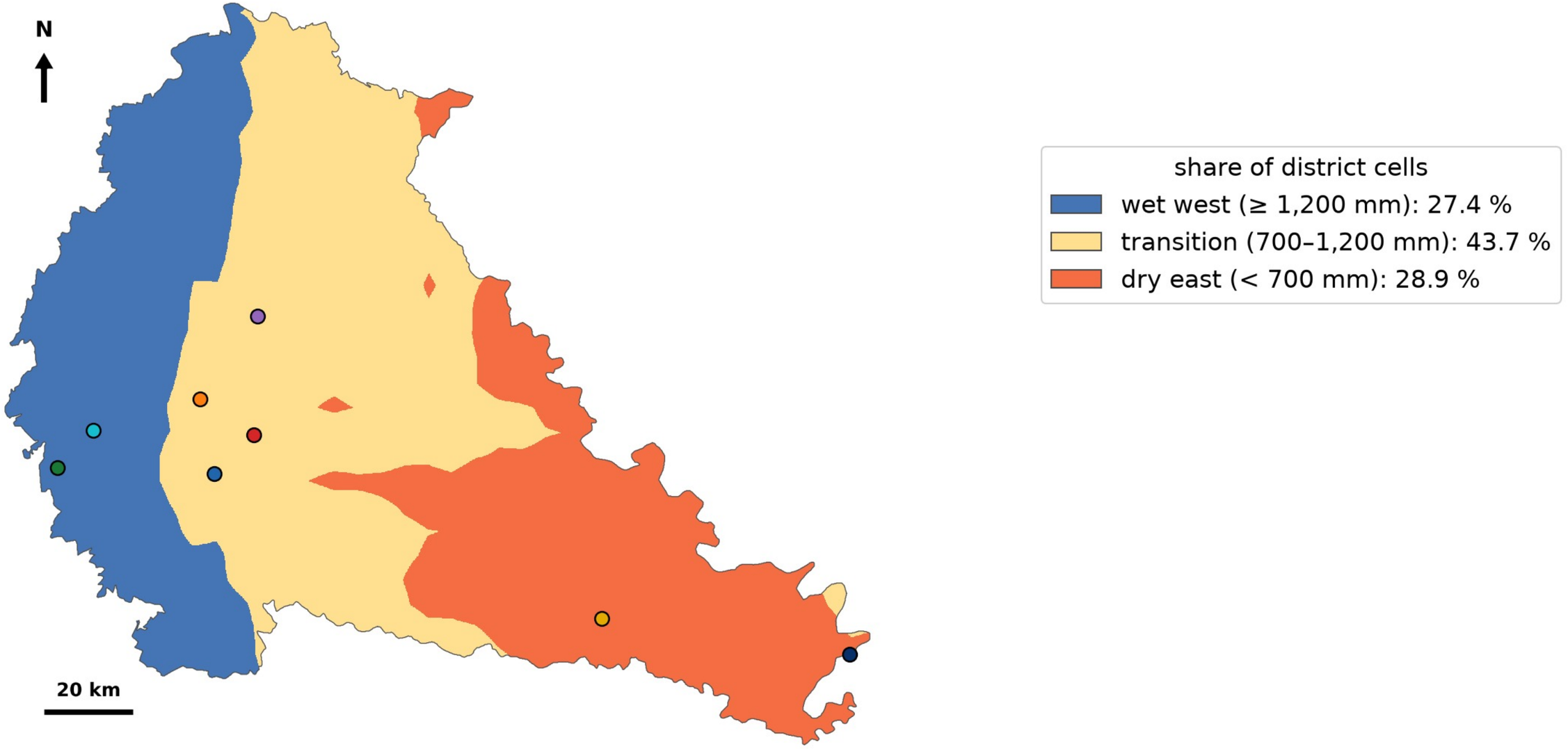
The district covers 17,380,093 grid cells × 900 m² (30 m × 30 m) = 15,642 km².
Grid: 6,546 × 5,624 cells in UTM zone 43N (EPSG:32643). Dashed lines: latitude and longitude every 0.5°.
Coloured dots: the places used throughout this tour (snapped to the nearest pixel that looks right in 2020).

Shape of the land: from the Western Ghats to the eastern plateau



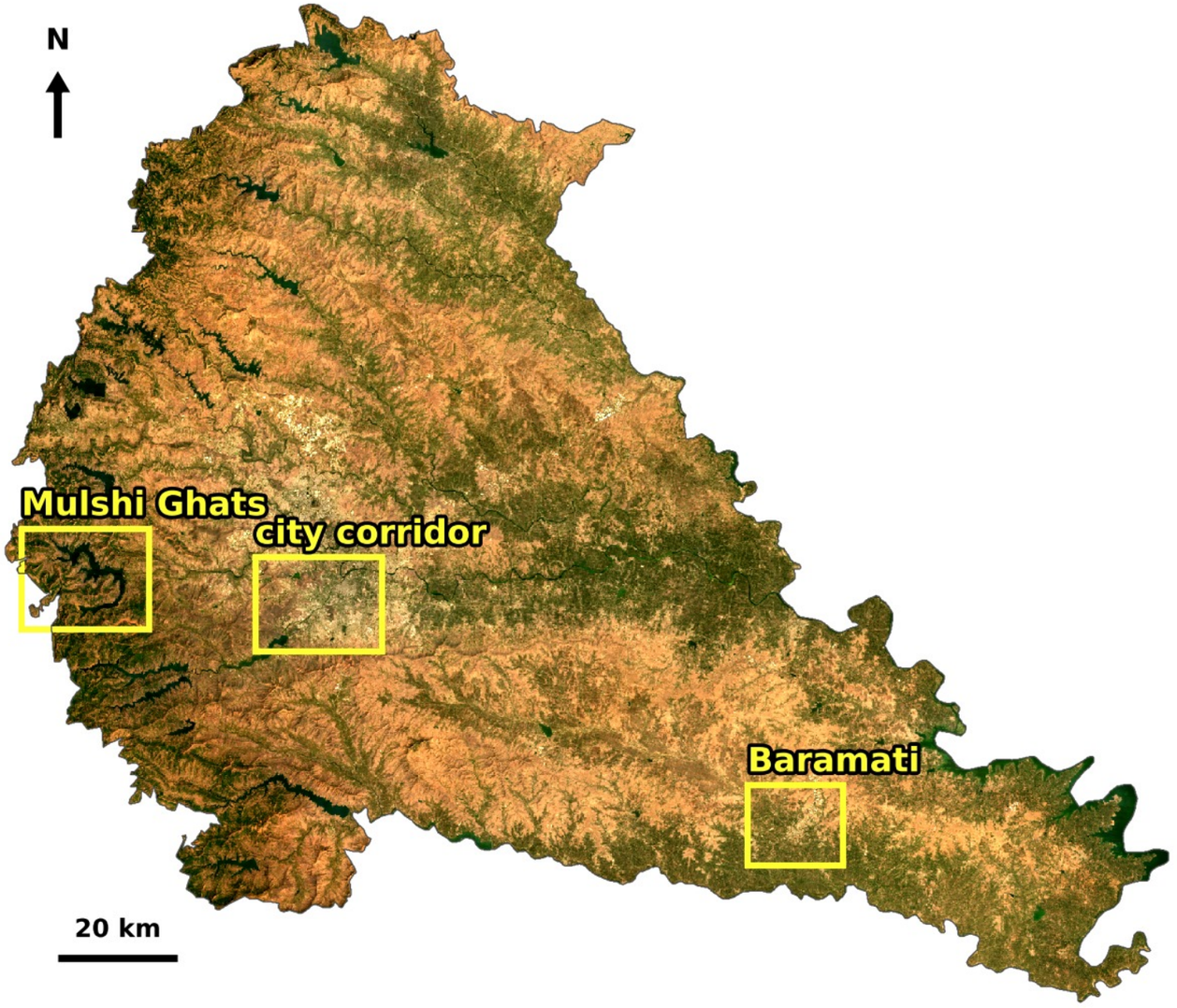
Elevation inside the district runs from 97 m to 1,476 m (Copernicus DEM, a surface model). Most of it lies between 493 and 1,006 m (2–98 %).
Median height falls from 725 m in the wet west to 630 m in the transition zone and 554 m in the dry east.
The highest ridges are in the west and south-west; from there the land falls gently towards the east.

Three rainfall regions: wet west, transition, dry east



Regions come from CHIRPS v2 rainfall: wet west ≥ 1,200 mm, transition 700–1,200 mm, dry east < 700 mm.
They are used to check whether results hold in every part of the district (e.g. leave-one-region-out tests).
Same three colours are used for the regions on every slide.

The three early test windows

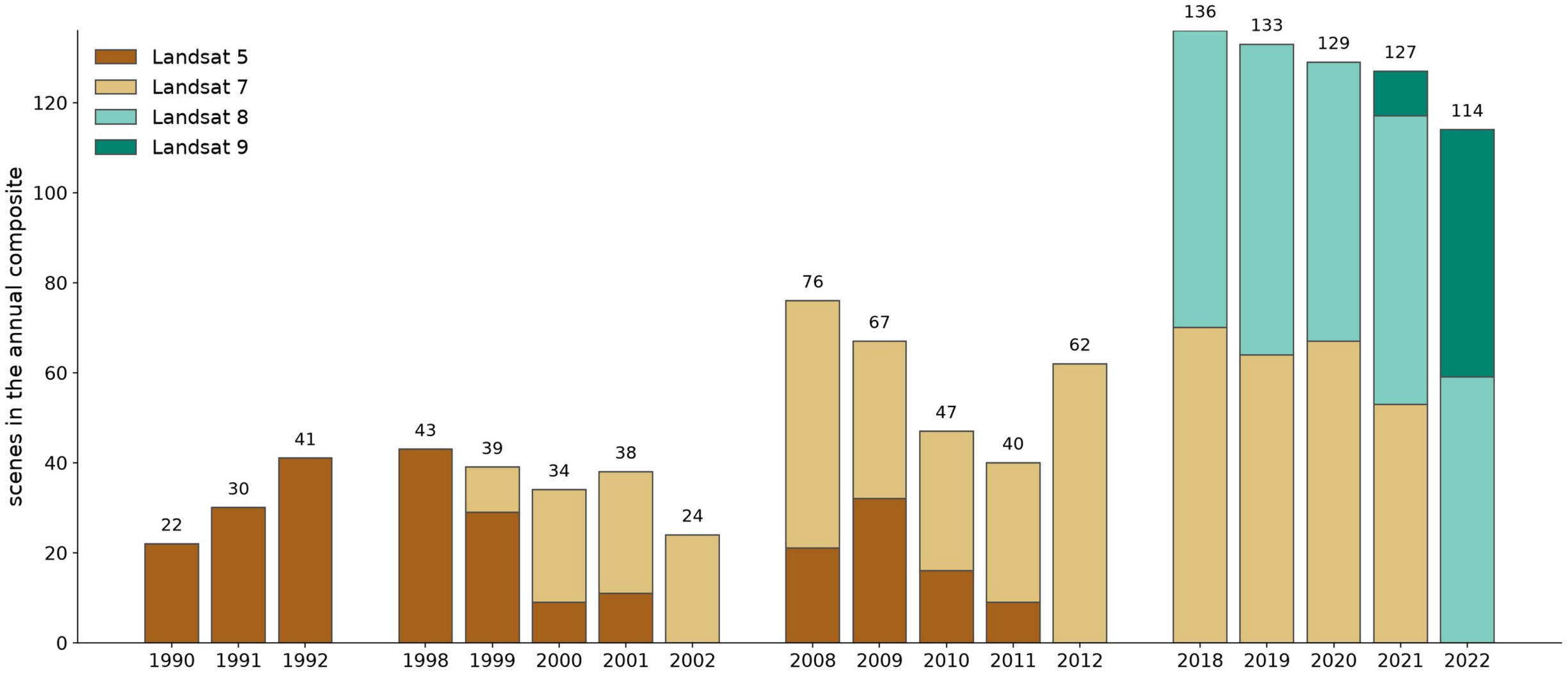


Yellow boxes: the windows (configs/aois.yaml), snapped outward to the 30 m grid.

Background: 2020 dry-season true colour (Landsat 7 + 8).

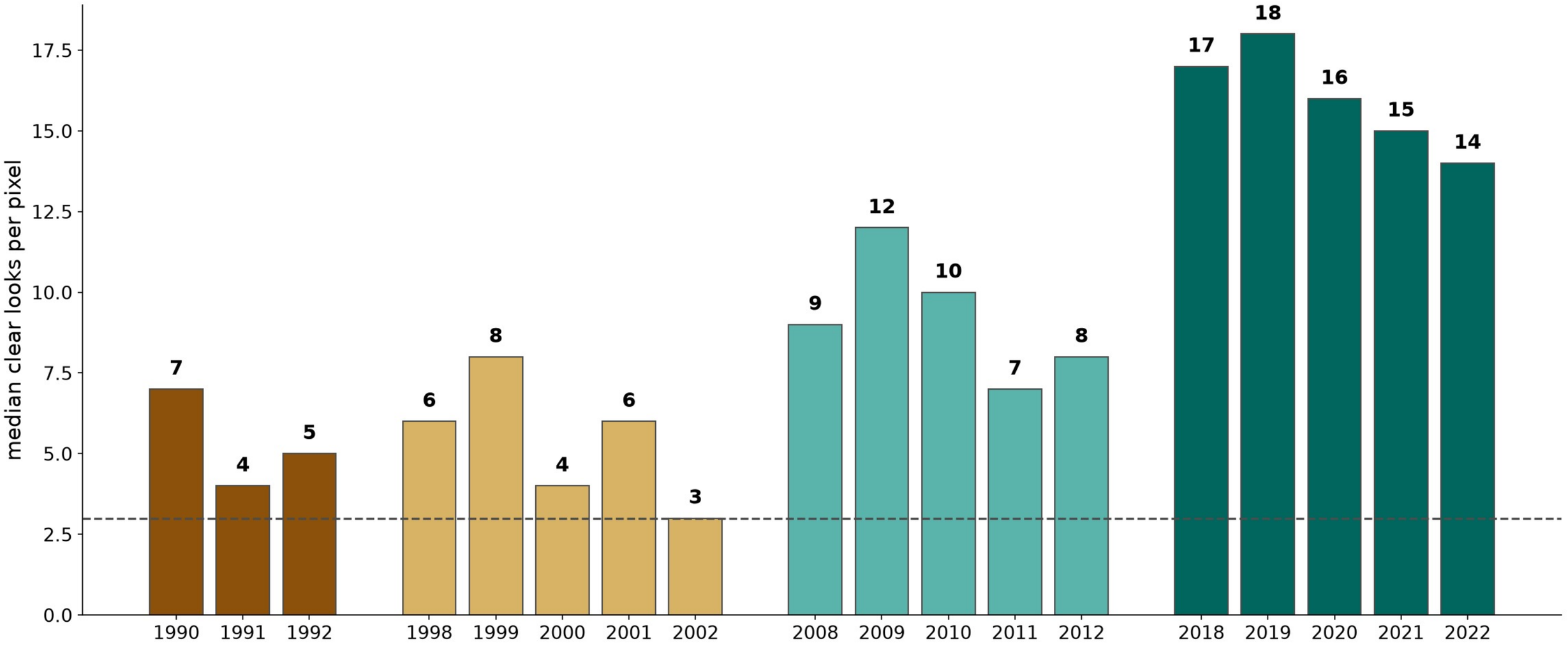
City corridor 21 × 16 km, Baramati farmland 16 × 13 km, Mulshi Ghats 21 × 17 km: methods were first tried here.
Together 1,003,433 cells = 5.8 % of the district's size (5.5 % of the district lies inside them; part of the Mulshi box is outside).
Validation (gold) points were kept away from them: the nearest is 2.01 km from a window.

Which satellite made each year



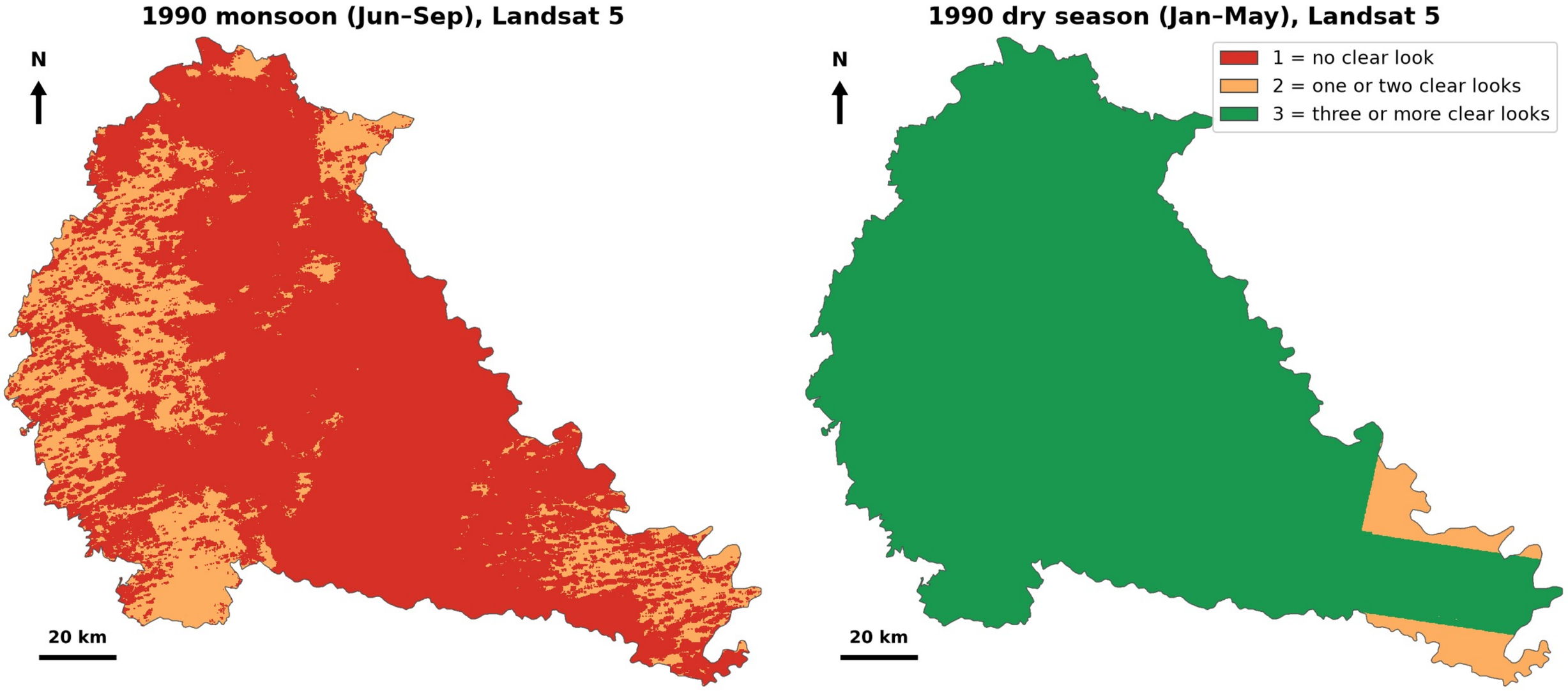
Each bar: the Landsat scenes that went into that year's annual composite, by satellite (from each product's metadata.json).
Early years rest on Landsat 5 alone; from 2018 two satellites overlap, so there are far more scenes.
18 years in four windows: 1990–92, 1998–2002, 2008–12, 2018–22.

How many clear pictures per pixel? (dry season, January–May)



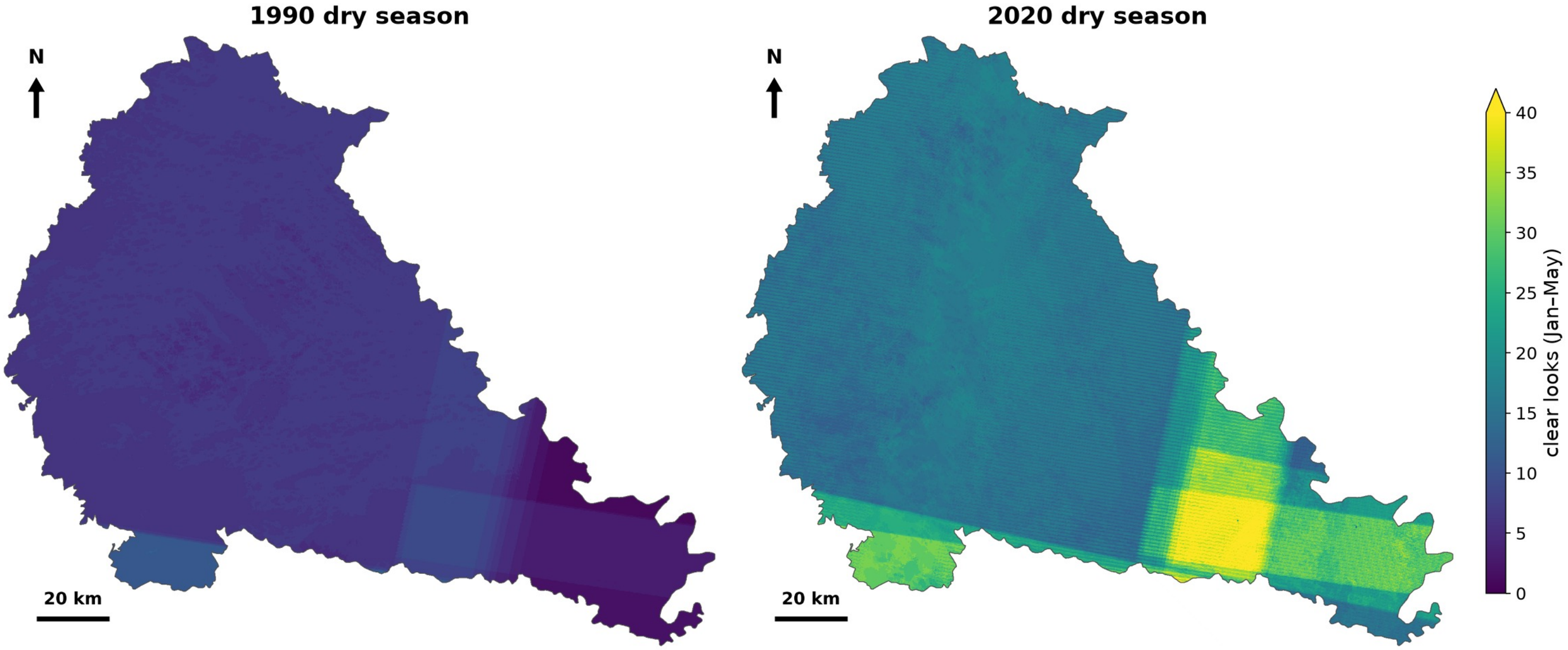
Median number of cloud-free looks per pixel in the dry-season composite (product metadata).
1990–2002: only 3–8 looks per pixel; 2018–22: 14–18. Fewer looks means each pixel's value is less certain.
Dashed line: 3 looks, the level the valid mask calls 'reliable'.

Why the dry season: the monsoon hides the ground



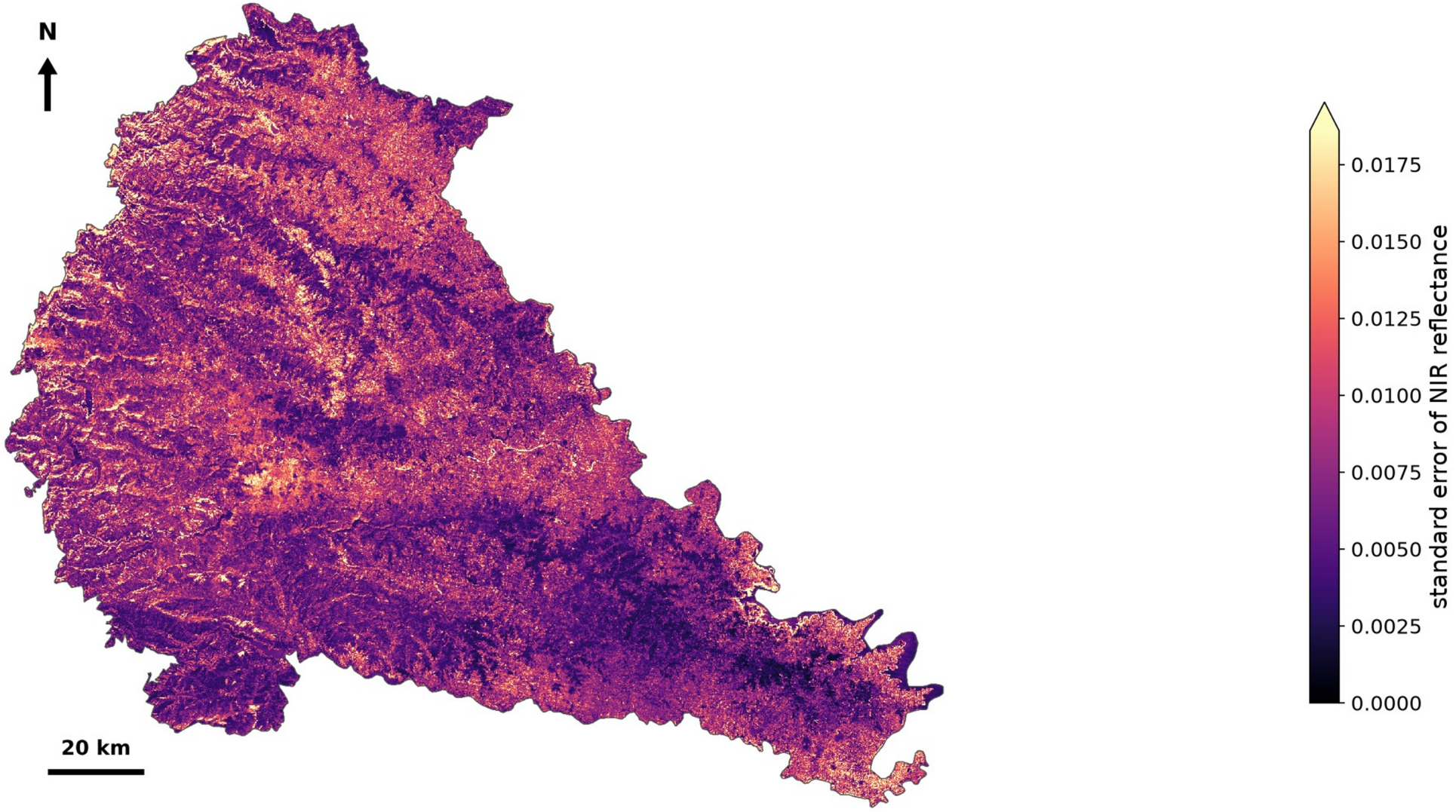
1990 monsoon (June–September): no clear look at all for 78 % of the district, and no pixel has 3 or more.
1990 dry season (January–May): 95.8 % of the district has 3 or more clear looks.
So the main record is built from dry-season pictures; other seasons are used only where they exist.

Clear looks per pixel: 1990 vs 2020 (dry season)



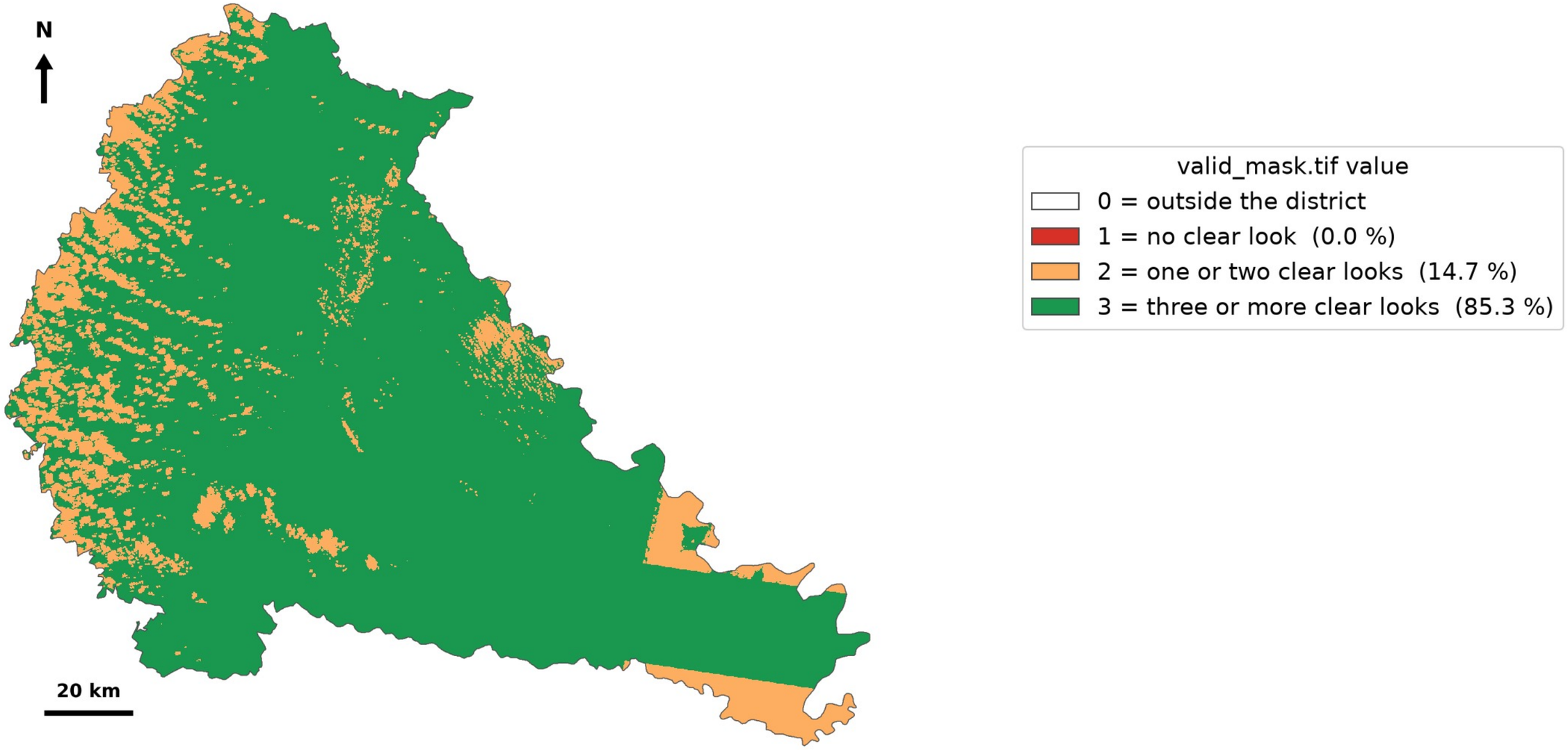
Each pixel's colour = how many cloud-free Landsat looks it had between January and May.
Median: 1990 = 7 looks (Landsat 5 only); 2020 = 16 looks (Landsat 7 + 8).
Brighter bands are where neighbouring satellite paths overlap; fine stripes in 2020 are Landsat 7 scan-line gaps.

How certain is each pixel? (near-infrared, dry season 2020)



Each pixel's value is the median of its clear looks. This map shows how much those looks disagreed (standard error of the median near-infrared reflectance; computed where there are 3 or more looks). District median: 0.0066. Dark = the looks agreed; bright = the surface or the looks varied during the season, so the value is less certain.

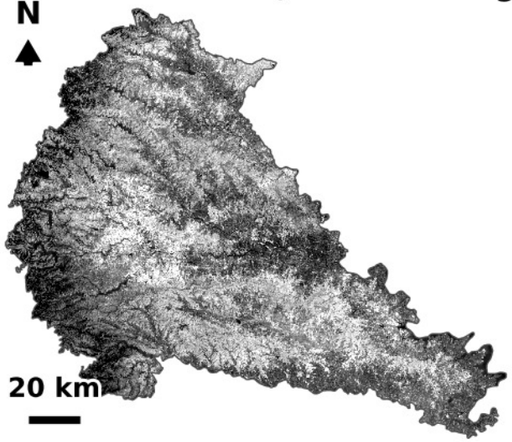
The valid mask: how many clear looks each pixel had



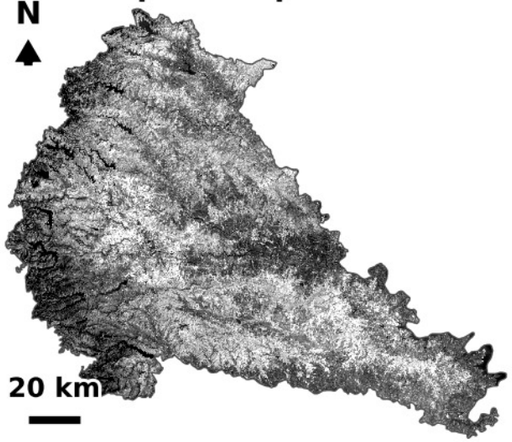
Every product carries this mask. 0 = outside the district (white); 1 = no clear look; 2 = one or two; 3 = three or more.
Shown: 2002 dry season, the thinnest year (median 3 looks; Landsat 7 only). Category 3 covers 85.3 %, category 2 14.7 %.
Pixels in category 2 still have a value, but it rests on one or two pictures and is flagged as less reliable.

Six colours of light: the bands of the 2020 dry-season composite

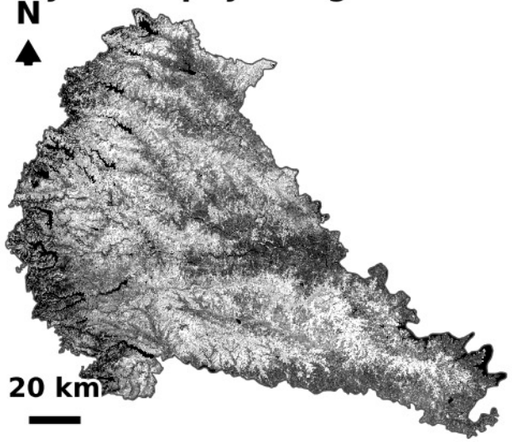
Blue
haze and shallow water; low over vegetation



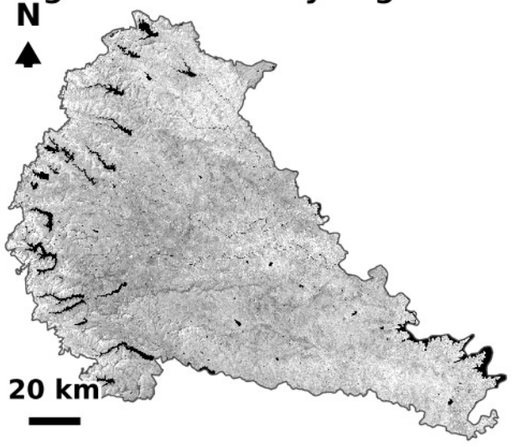
Green
the visible peak of plant reflectance



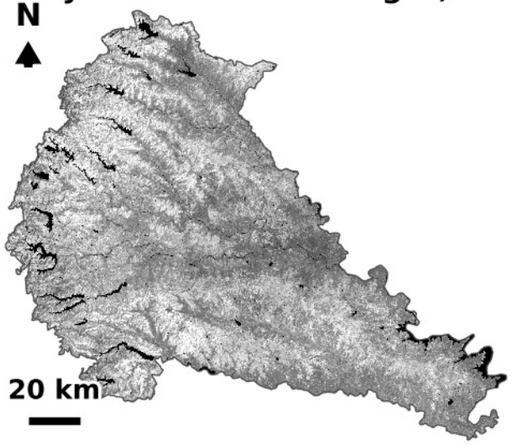
Red
absorbed by chlorophyll: vegetation looks dark



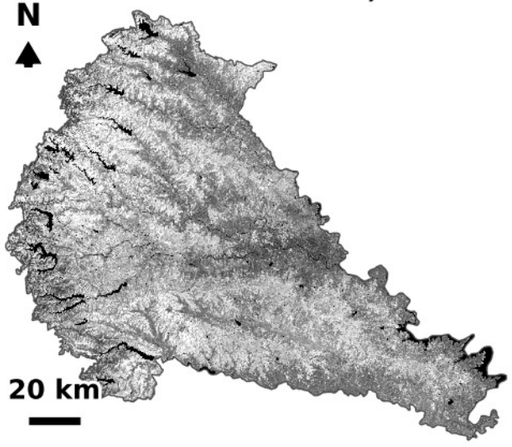
Near-infrared
bright over healthy vegetation



Shortwave-IR 1
moisture: dry soil and roofs bright, water black

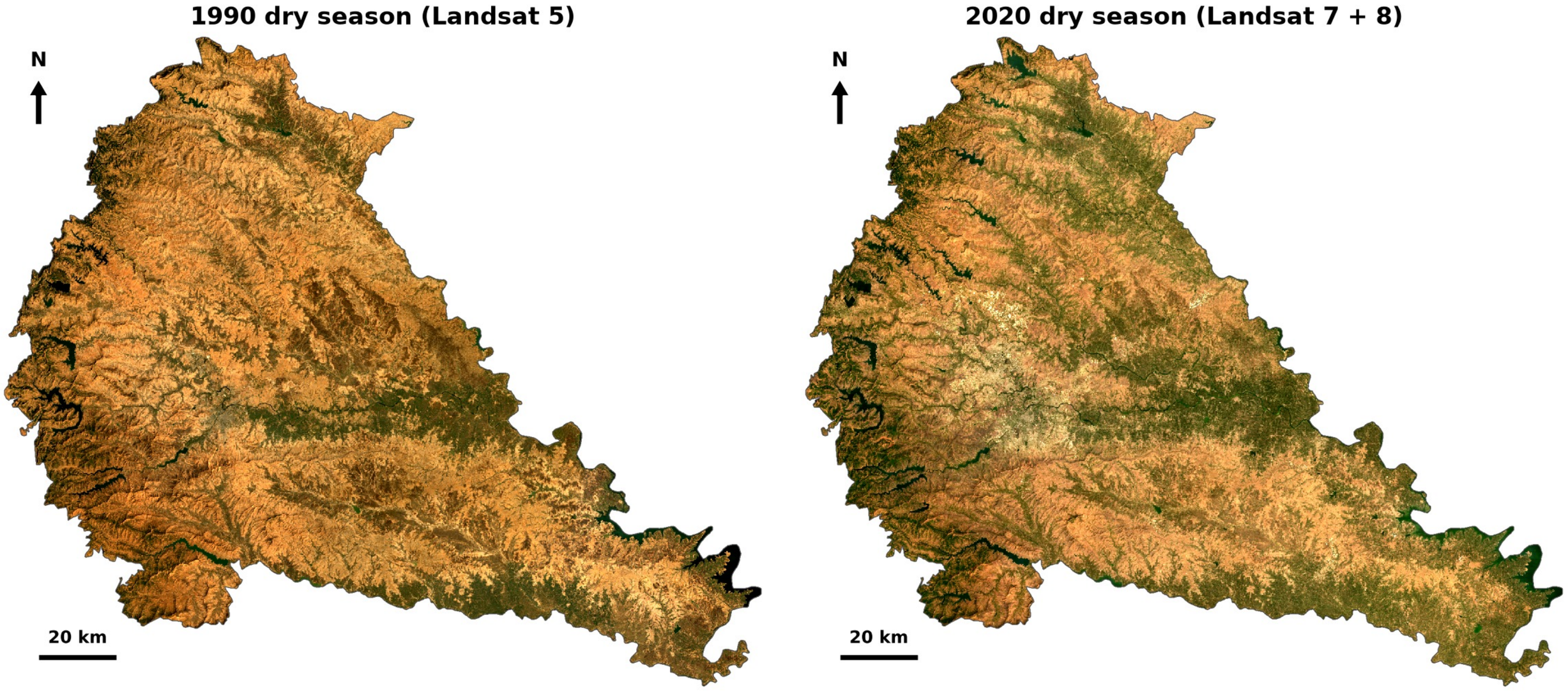


Shortwave-IR 2
dry surfaces and minerals; water black



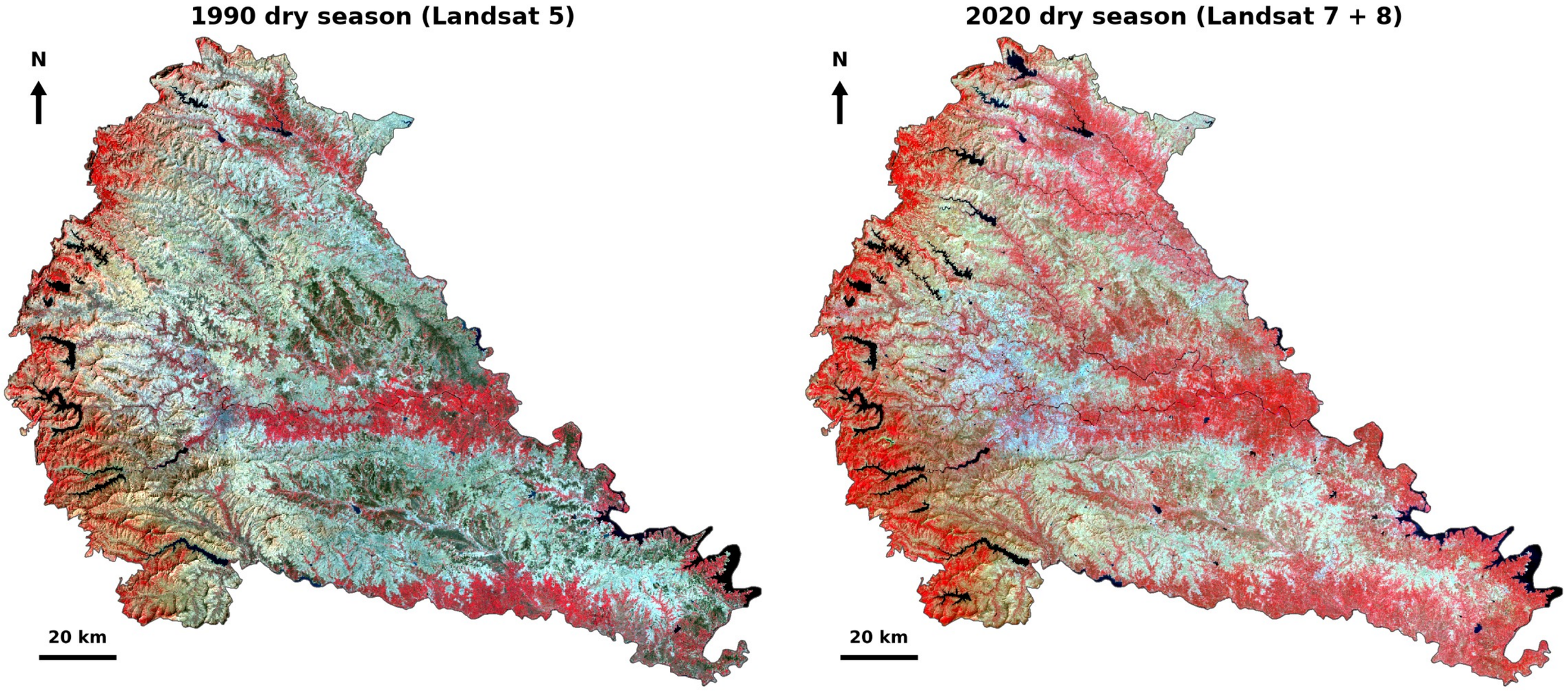
Each map is one band of optical_composite.tif (surface reflectance; 2–98 % grey stretch per band; white outside the district).
Together the six bands let us tell water, vegetation, soil and built surfaces apart.

The whole district in true colour: 1990 vs 2020



Natural colours from the red, green and blue bands of the dry-season composites (2–98 % stretch per image, one range for all three bands).
Dark green = trees and green crops; orange-brown = dry soil and grass; grey-white = roofs, roads and bare rock; black = water.
Part of any 1990-vs-2020 difference may come from the different sensors and from different rainfall years, not only from land change.

False colour (near-infrared, red, green): red = vegetation

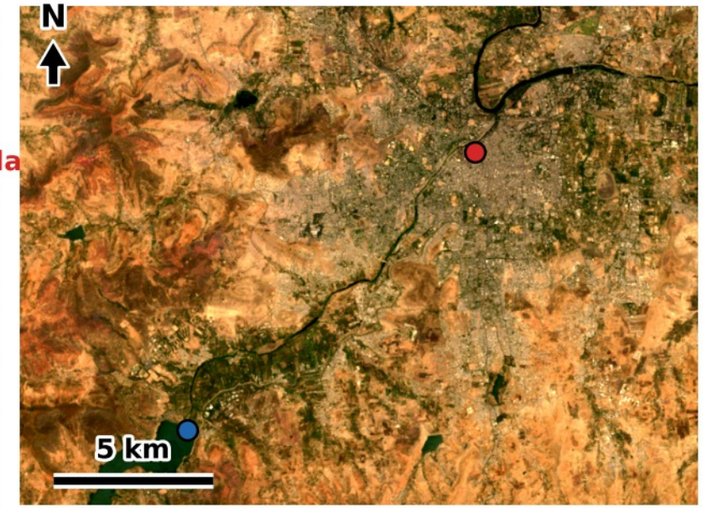


Near-infrared is shown as red, so living plants glow red; bare soil looks blue-grey or tan, water black, roofs and roads grey-blue.
Dry-season composites, 2–98 % stretch per band and image.
Part of any 1990-vs-2020 difference may come from the different sensors and from different rainfall years, not only from land change.

Pune city corridor, 1990 · 2000 · 2010 · 2020 (full 30 m)

1990 · Landsat 5

2000 · Landsat 5 + 7



2010 · Landsat 5 + 7

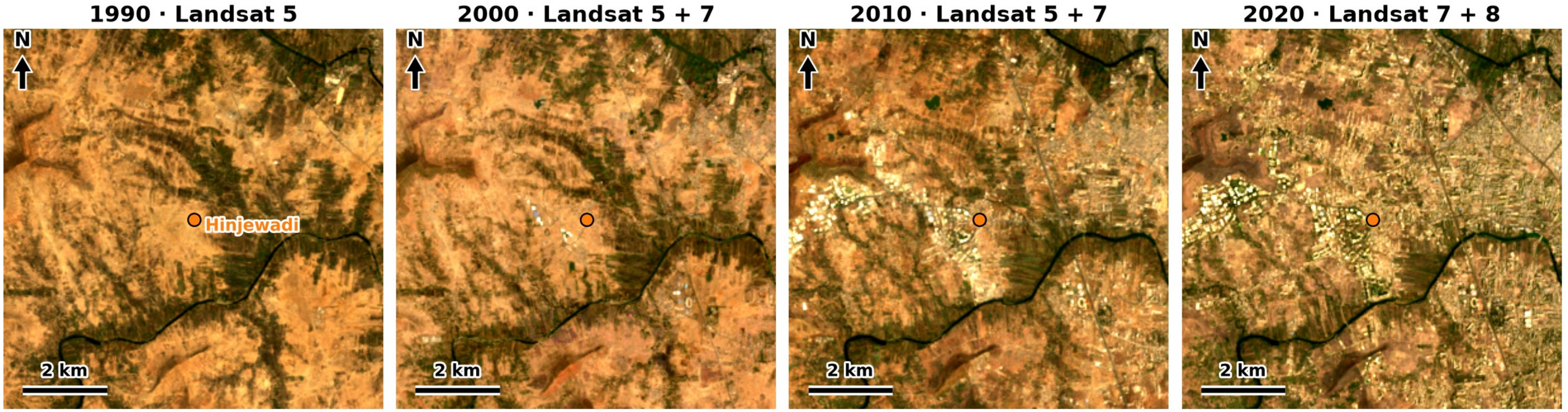
2020 · Landsat 7 + 8

What to look for

- the grey-white city spreading outwards from the old core (red dot) between 1990 and 2020
- new grey areas replacing orange-brown fields and hills
- the reservoir (blue dot) at the same place in every year

The early test window (21 × 16 km): Khadakwasla reservoir (bottom left) and the Mutha river running north-east through the city.
Grey-white = roofs and roads; dark green = trees and green crops; orange-brown = dry fields and hills; black = water.
Each picture: dry-season median composite, 2–98 % stretch per picture. Colours also differ between sensors.

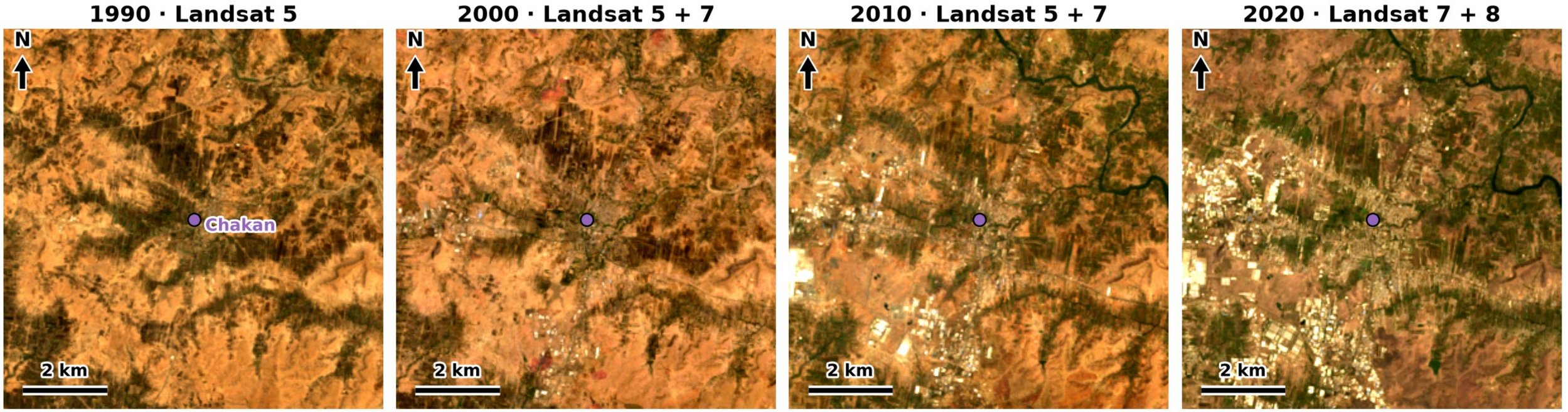
Hinjewadi IT park, 1990 · 2000 · 2010 · 2020 (full 30 m)



Look for: bright white blocks (large roofs) appearing around and west of the dot by 2010 and filling in by 2020.

9 × 9 km around the Hinjewadi pixel (orange dot), dry-season true colour, 2–98 % stretch per picture.
Grey-white = roofs, roads and cleared ground; orange-brown = dry fields; dark green = trees and green crops.
Colours also differ between sensors (Landsat 5 in 1990; Landsat 7 + 8 in 2020).

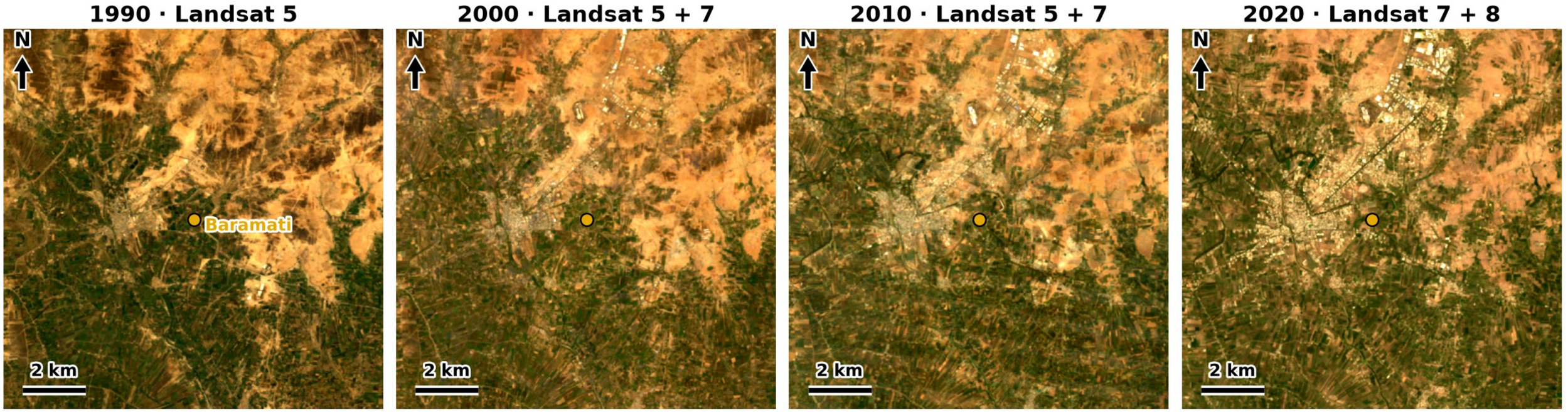
Chakan industrial area, 1990 · 2000 · 2010 · 2020 (full 30 m)



Look for: large bright rectangles appearing by 2010 and spreading by 2020, mostly west and south of the dot.

9 × 9 km around the Chakan pixel (purple dot), dry-season true colour, 2–98 % stretch per picture.
Large bright rectangles are typical of factory roofs and yards; orange-brown = dry fields.
Colours also differ between sensors (Landsat 5 in 1990; Landsat 7 + 8 in 2020).

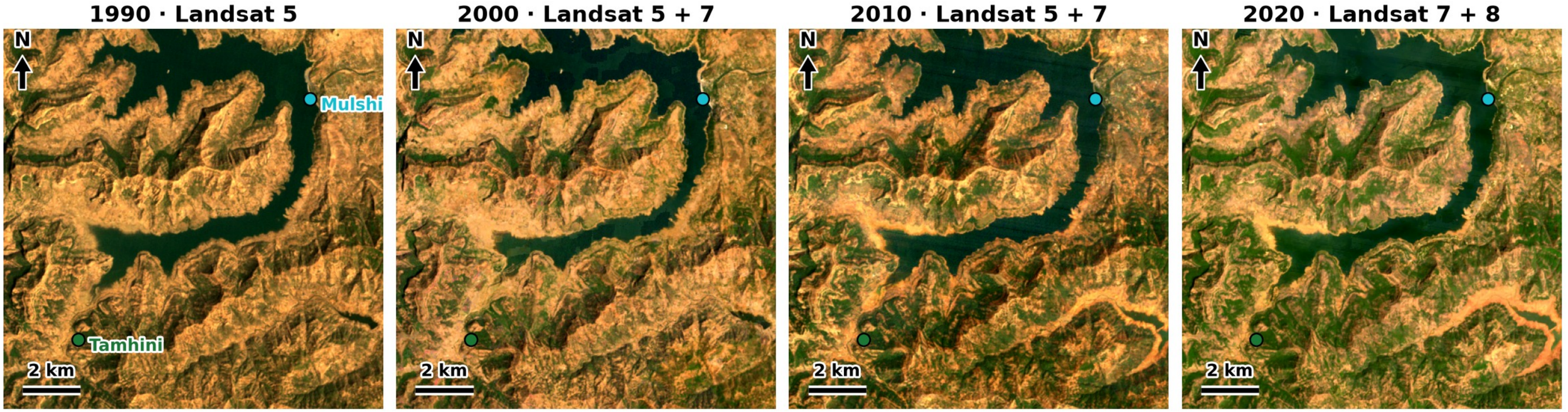
Baramati farmland, 1990 · 2000 · 2010 · 2020 (full 30 m)



Look for: the grey town west of the dot growing outwards; green (irrigated) fields present in all four years.

12 × 12 km around a green-field pixel east of Baramati town (gold dot), dry-season true colour.
In the dry season, green fields need irrigation; brown fields are fallow or harvested. The town is the grey patch to the west.
Colours also differ between sensors (Landsat 5 in 1990; Landsat 7 + 8 in 2020).

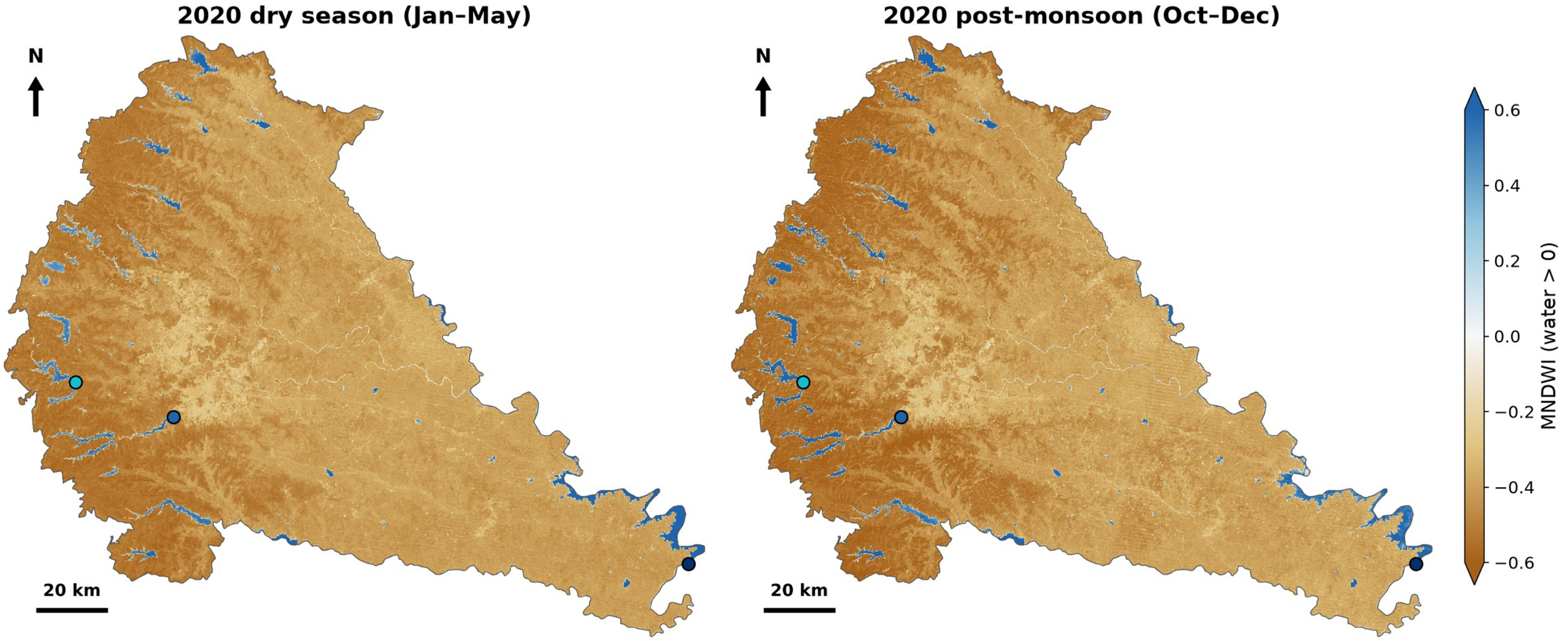
Mulshi reservoir and Tamhini Ghat forest, 1990 · 2000 · 2010 · 2020



Look for: a second water body (bottom right) much larger in 2010 and 2020; fine stripes in 2010 = Landsat 7 gaps.

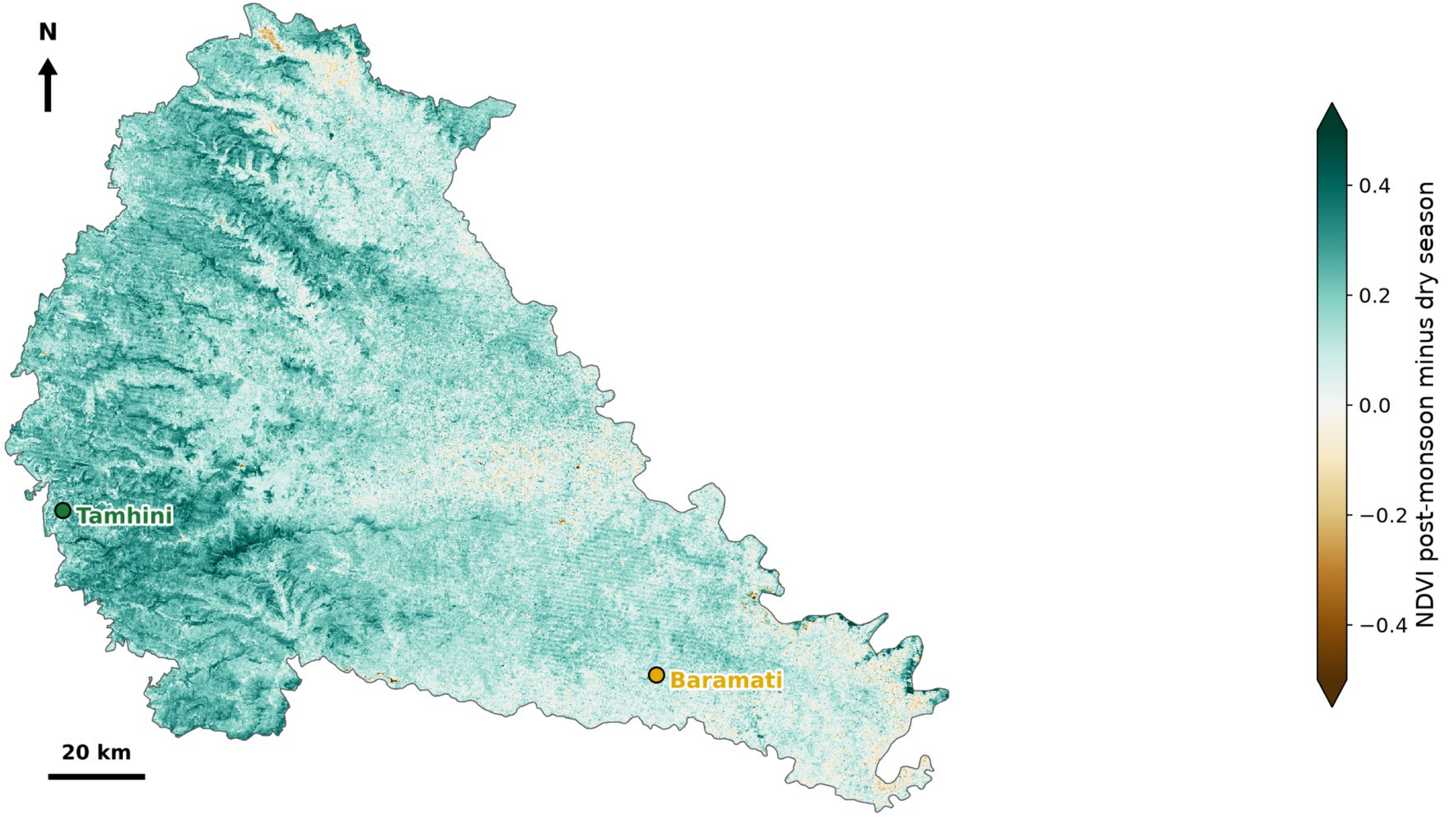
13.1 × 13.1 km at full 30 m around Mulshi reservoir (cyan dot) and a Tamhini forest pixel (green dot); dry-season true colour.
Dark green = forest; orange-brown = dry grass and rock on the hill slopes; near-black = reservoir water.
Each picture is a January–May median, so the water edge shows that year's dry-season level.

Water: dry season vs after the monsoon, 2020 (MNDWI)



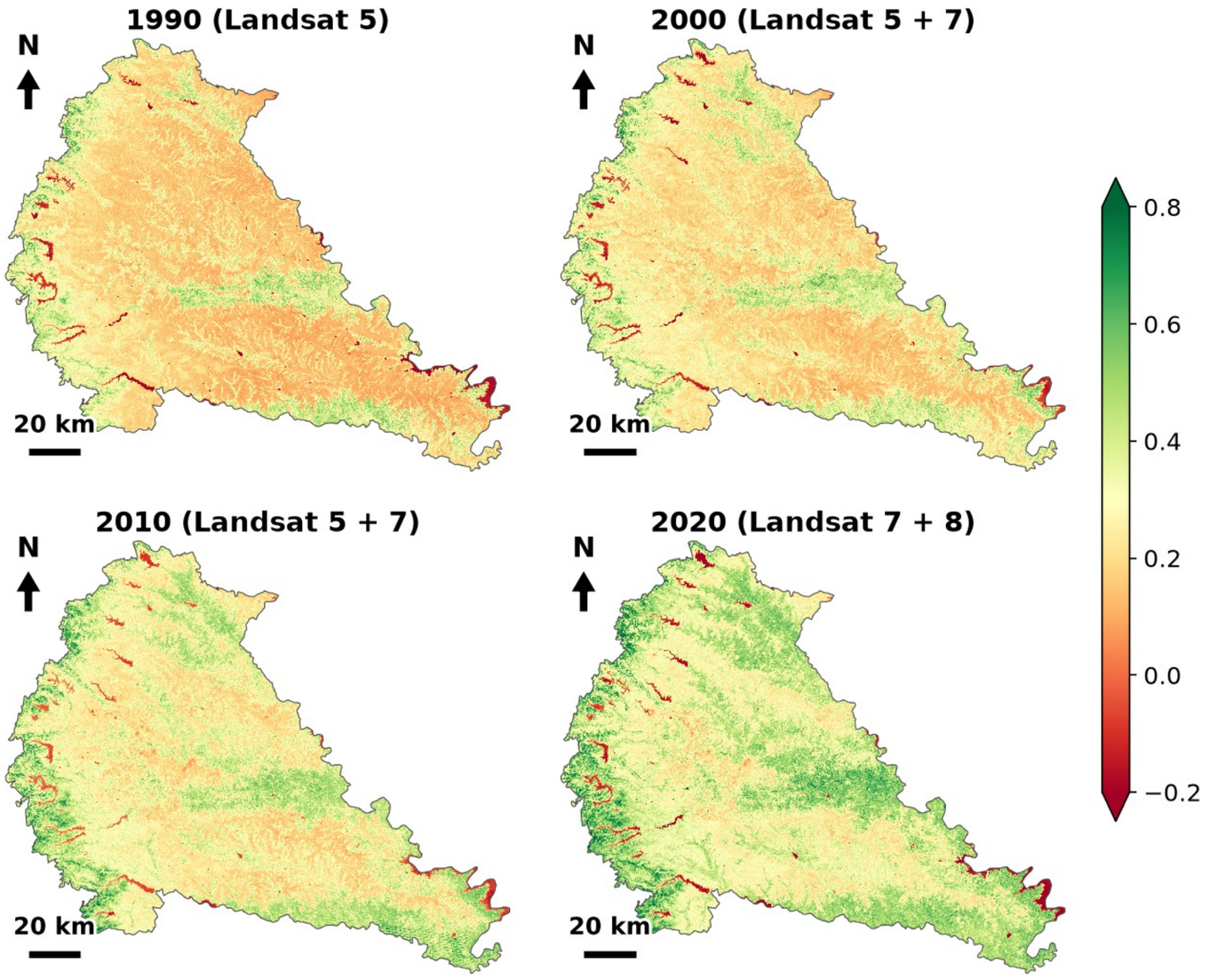
MNDWI compares green and shortwave-infrared light; open water is above 0 (blue), land below 0 (brown).
Area with MNDWI above 0 (quarter-size maps): about 411 km² in the dry season vs 496 km² after the monsoon.
Reservoirs are fuller after the monsoon: compare Khadakwasla (blue), Mulshi (cyan) and the Ujani backwater (navy).

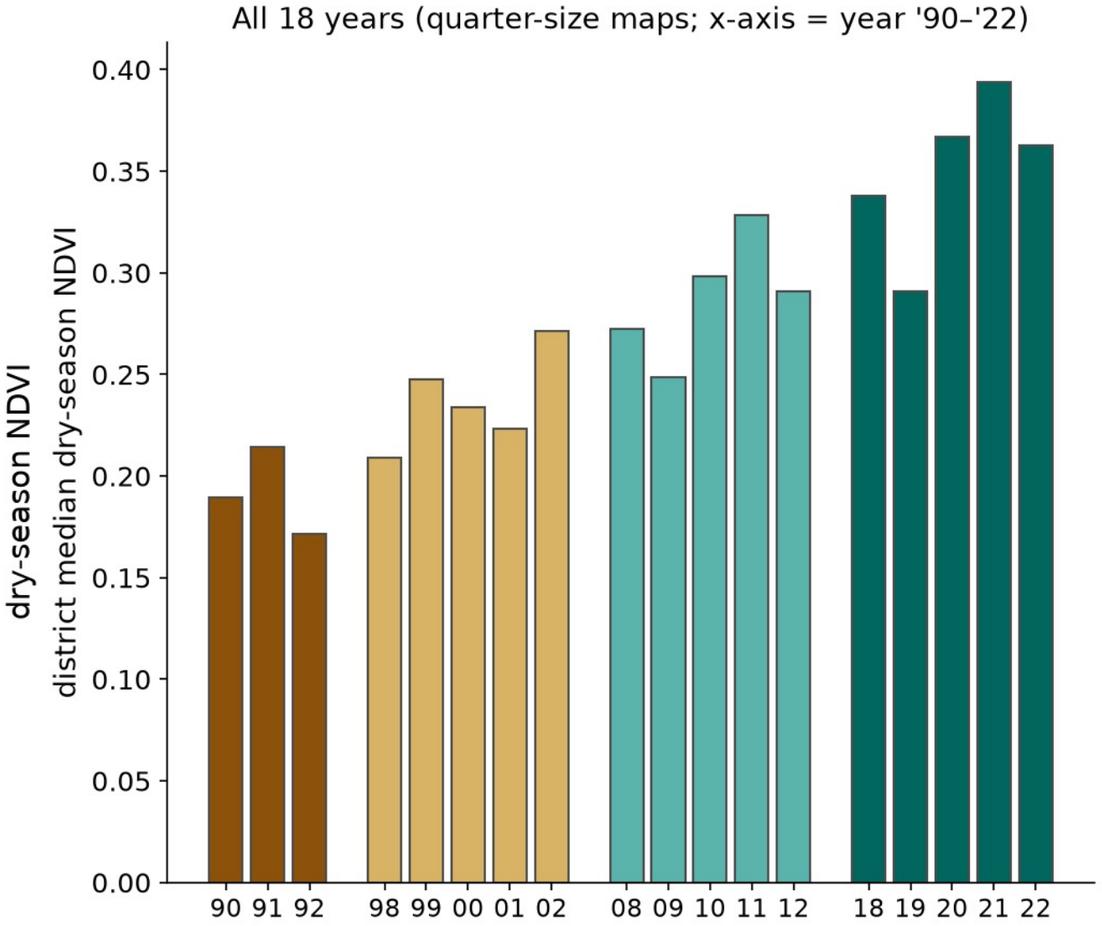
Seasonal greenness swing, 2020: green only after rain, or green all year?



Colour = NDVI after the monsoon (Oct–Dec 2020) minus NDVI in the dry season (Jan–May 2020).
Green = much greener after the rains (typical of rain-fed fields and grass). Near 0 = similar in both seasons (irrigated crops, forest, or bare).
Examples (3 × 3 pixels): Baramati green field dry 0.60 → post 0.61; Tamhini forest dry 0.69 → post 0.83.

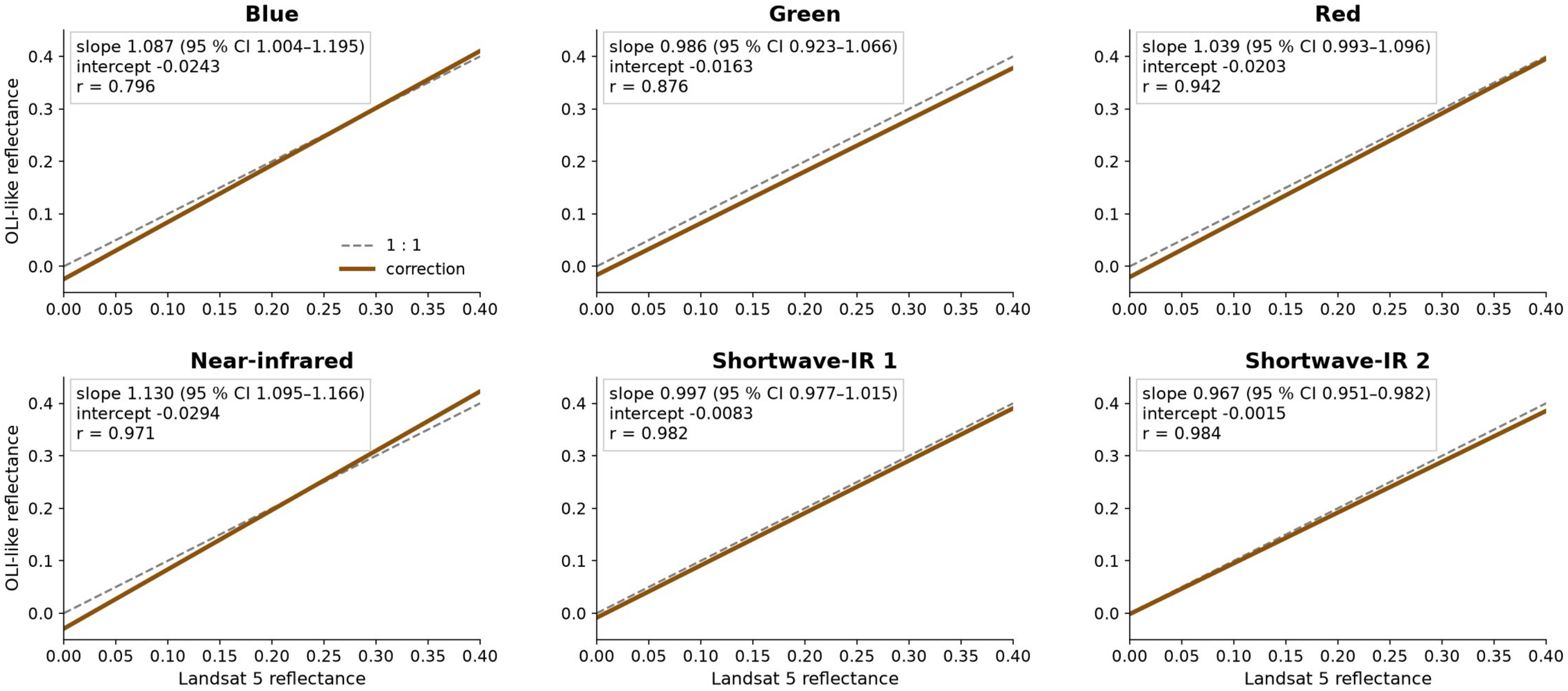
Dry-season greenness (NDVI) in four epochs





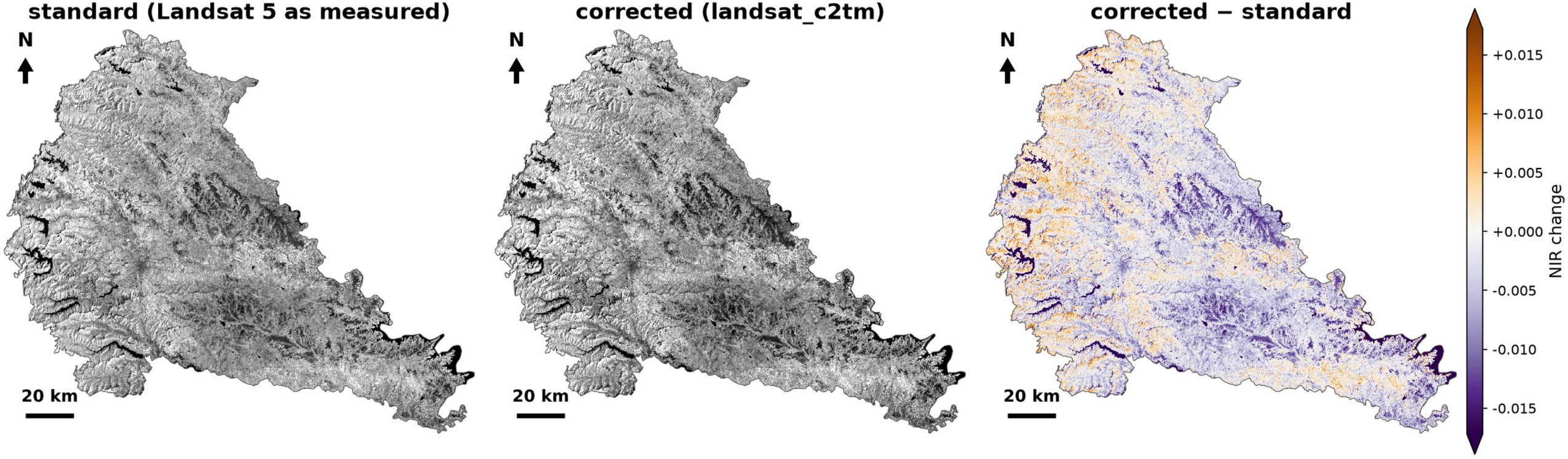
NDVI = (near-infrared − red) / (near-infrared + red): about 0 for bare ground, higher for green plants. One shared colour scale.
District medians: 1990 0.19, 2000 0.23, 2010 0.30, 2020 0.37.
Part of any 1990-vs-2020 difference may come from the different sensors and from different rainfall years, not only from land change.

Why sensors differ: the Landsat 5 → OLI correction (c2_tm_pif_v1)



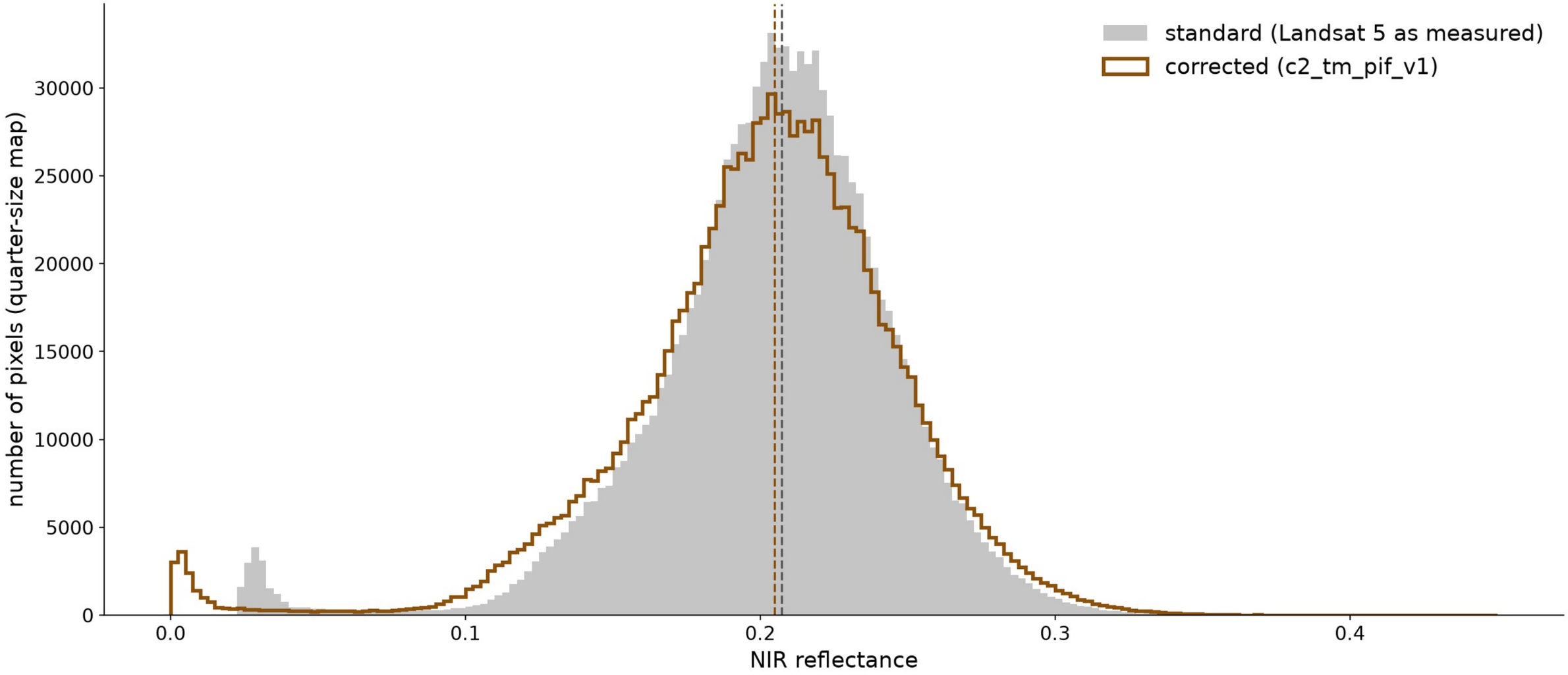
Each line maps Landsat 5 (TM) reflectance to the OLI-like scale of the cube: corrected = slope × TM + intercept. Dashed = no change (1 : 1).
Fitted on surfaces assumed stable (pseudo-invariant): 110,000 water, 48,012 forest and 37,445 impervious cells, 45 image pairs.
Used only for the C2 test ('landsat_c2tm' products); the standard products keep Landsat 5 values as they are.

1990 near-infrared: standard vs corrected



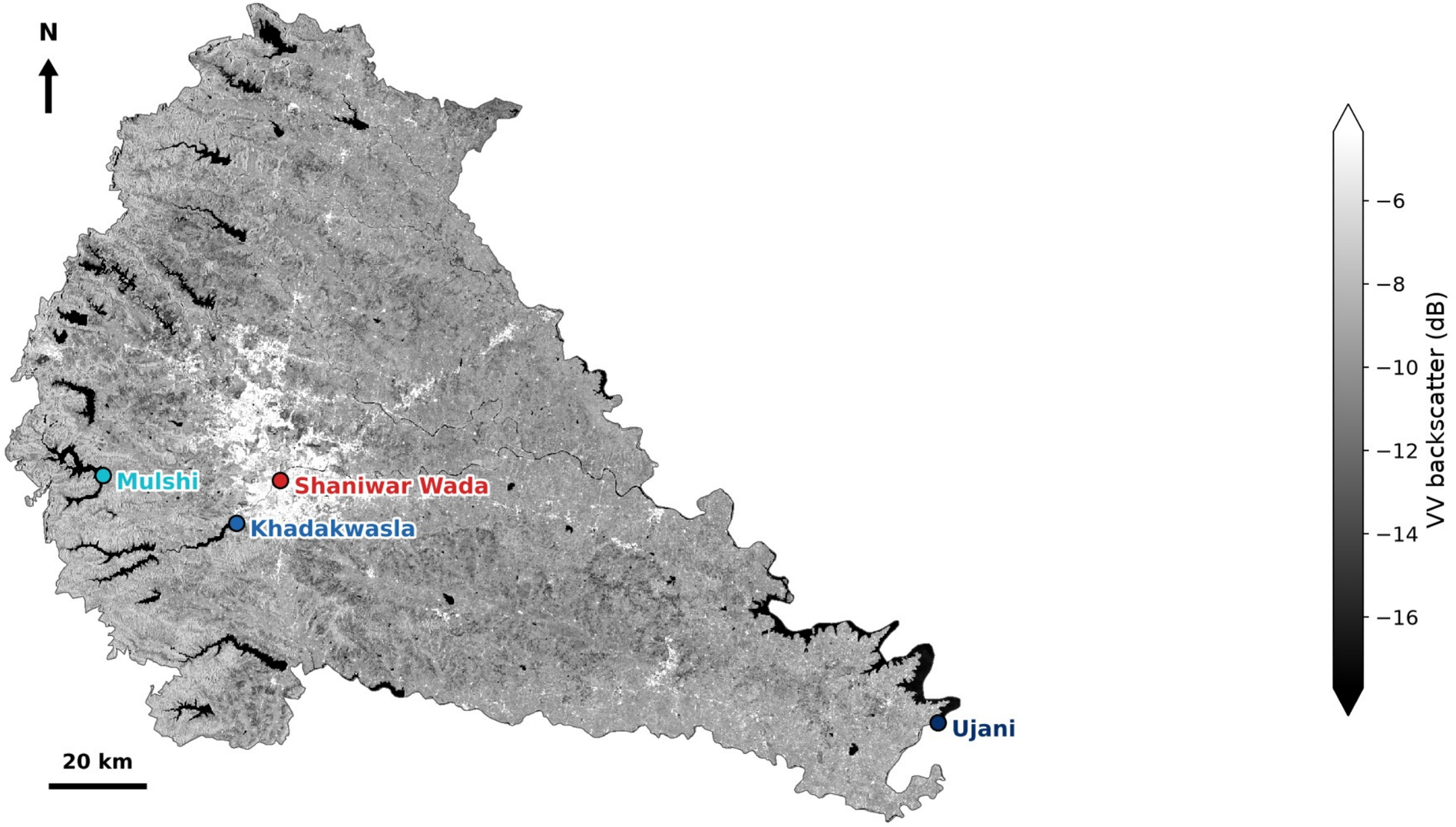
Left: 1990 dry-season NIR as Landsat 5 measured it (standard cube). Middle: the same after the c2_tm_pif_v1 correction. Same grey scale. Right: corrected minus standard. Dark land is lowered a little, bright land raised (slope 1.130, intercept −0.029); district median change -0.0024. The correction changes values, not the picture's pattern: it is a straight-line rescaling, the same for every pixel.

1990 near-infrared values before and after the correction



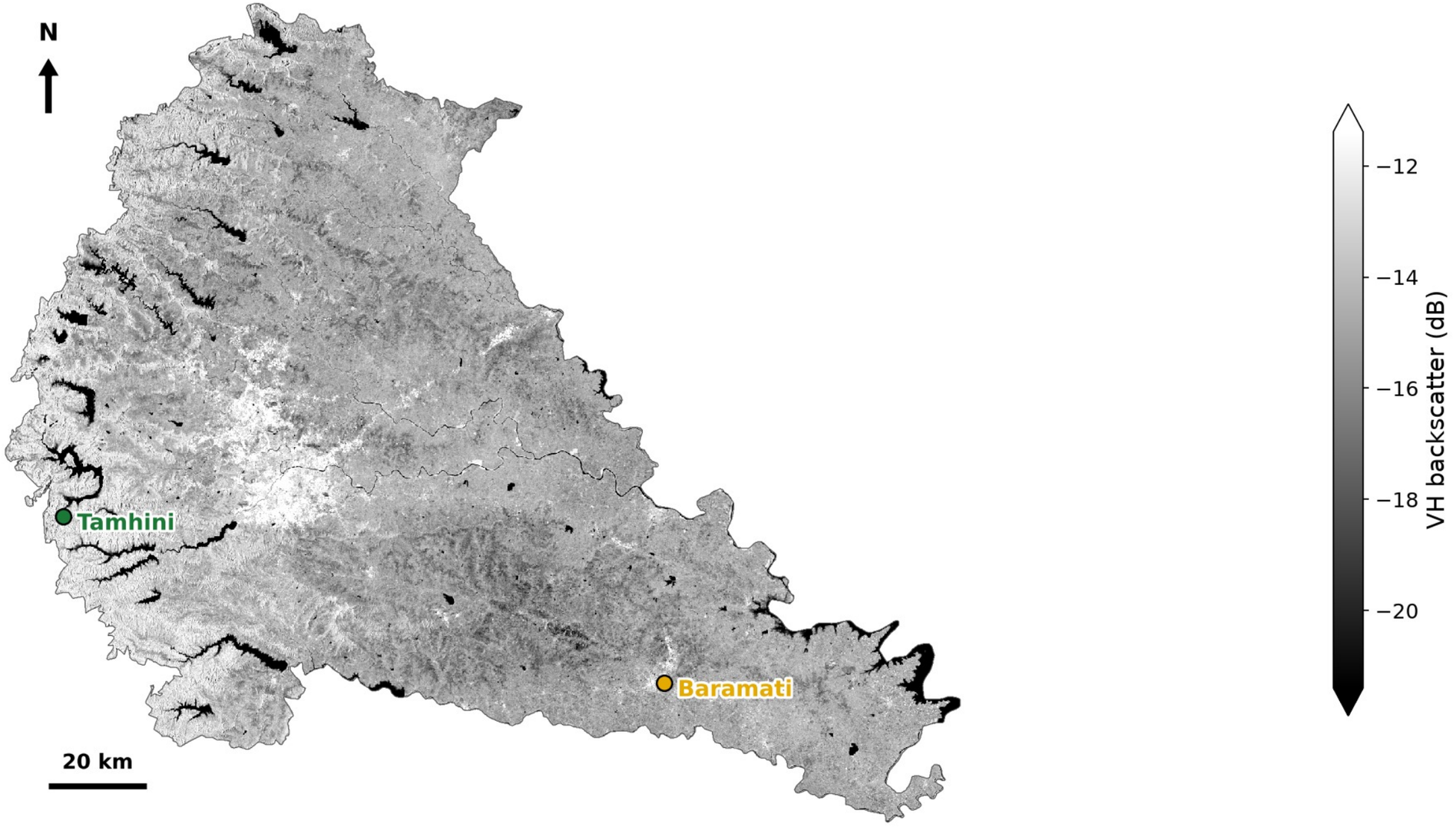
Histograms of 1990 dry-season NIR reflectance over all district pixels (quarter-size maps). Medians: standard 0.2073, corrected 0.2048.
Because the slope is above 1, the corrected values are spread wider; the intercept shifts them down slightly.
Whether this makes 1990 maps better is exactly what test C2 will check against labelled points; no result exists yet.

Radar: Sentinel-1 VV, dry season 2020



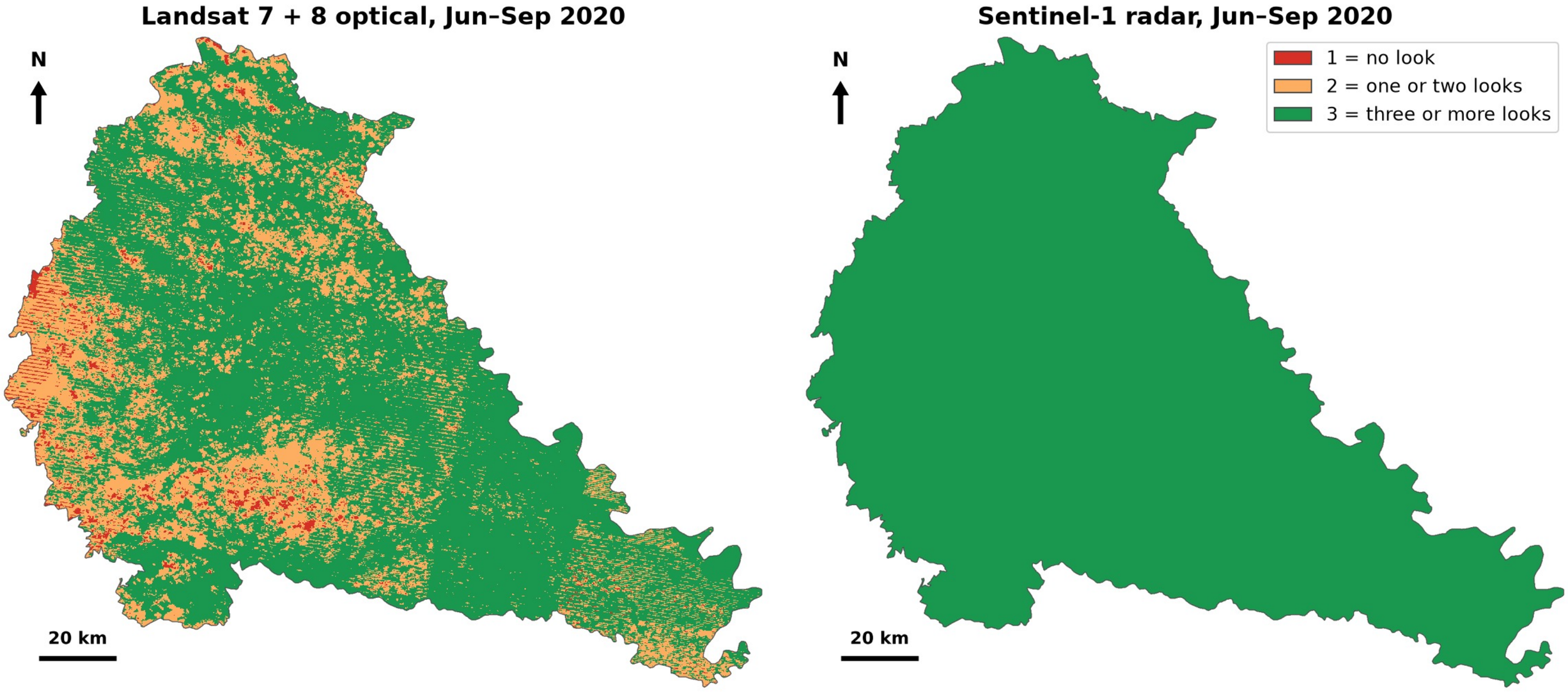
Median VV backscatter of 44 radar scenes (Sentinel-1a: 40, Sentinel-1b: 4), 2020-01-05 to 2020-05-28. Grey scale 2–98 % (-17.7 to -4.4 dB).
Radar sends its own microwaves and sees through cloud. Smooth water reflects them away, so reservoirs are dark;
buildings and slopes facing the satellite send a lot back, so the city and hill sides are bright.

Radar in the monsoon: Sentinel-1 VH, June–September 2020



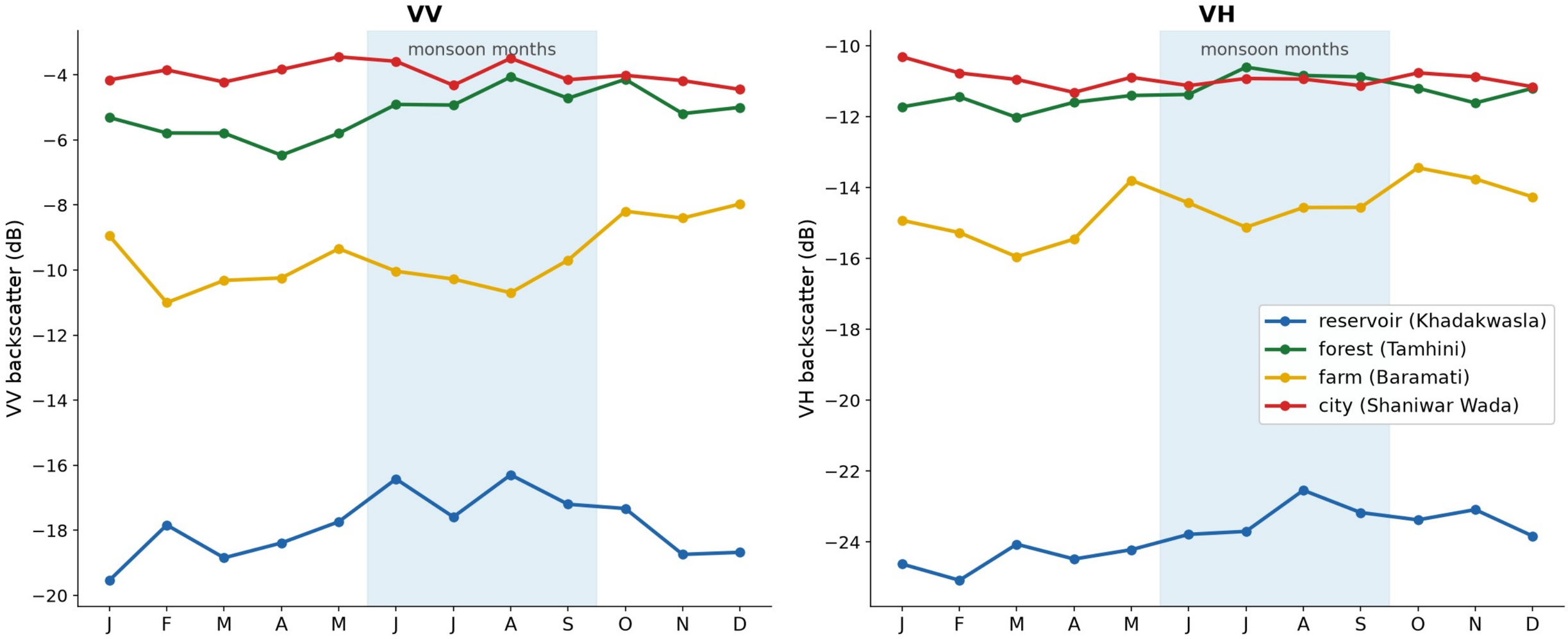
Median VH backscatter, 2020-06-02 to 2020-09-25 (40 radar scenes). VH responds to leaves and stems, so crops and forest are brighter.
Radar had a median of 10 looks per pixel in these months; Landsat had a median of only 3 clear looks (cloud).
Grey scale 2–98 % (-21.4 to -11.4 dB). Dark = water and bare, smooth ground.

Same season, two sensors: monsoon 2020 (June–September)



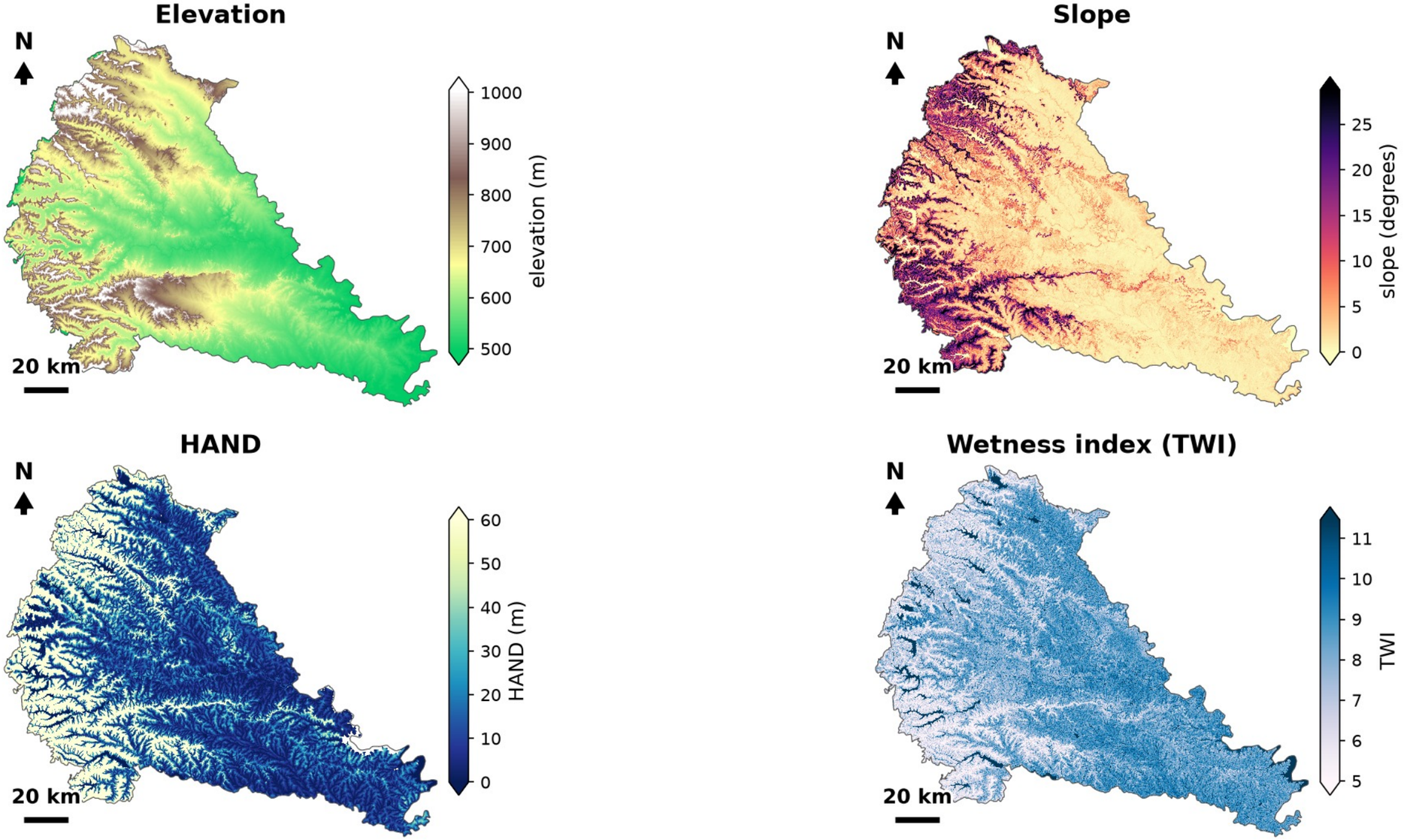
Landsat (left): at least one clear look for 97.9 % of the district, but three or more only for 69.4 %; 2.1 % had none.
Sentinel-1 radar (right): three or more looks for 100 % of the district, because cloud does not stop radar.
Two Landsat satellites gave far more monsoon looks in 2020 than one did in 1990: then (slide 8) 78 % of the district had no clear look at all.

Radar month by month, 2020: four kinds of place



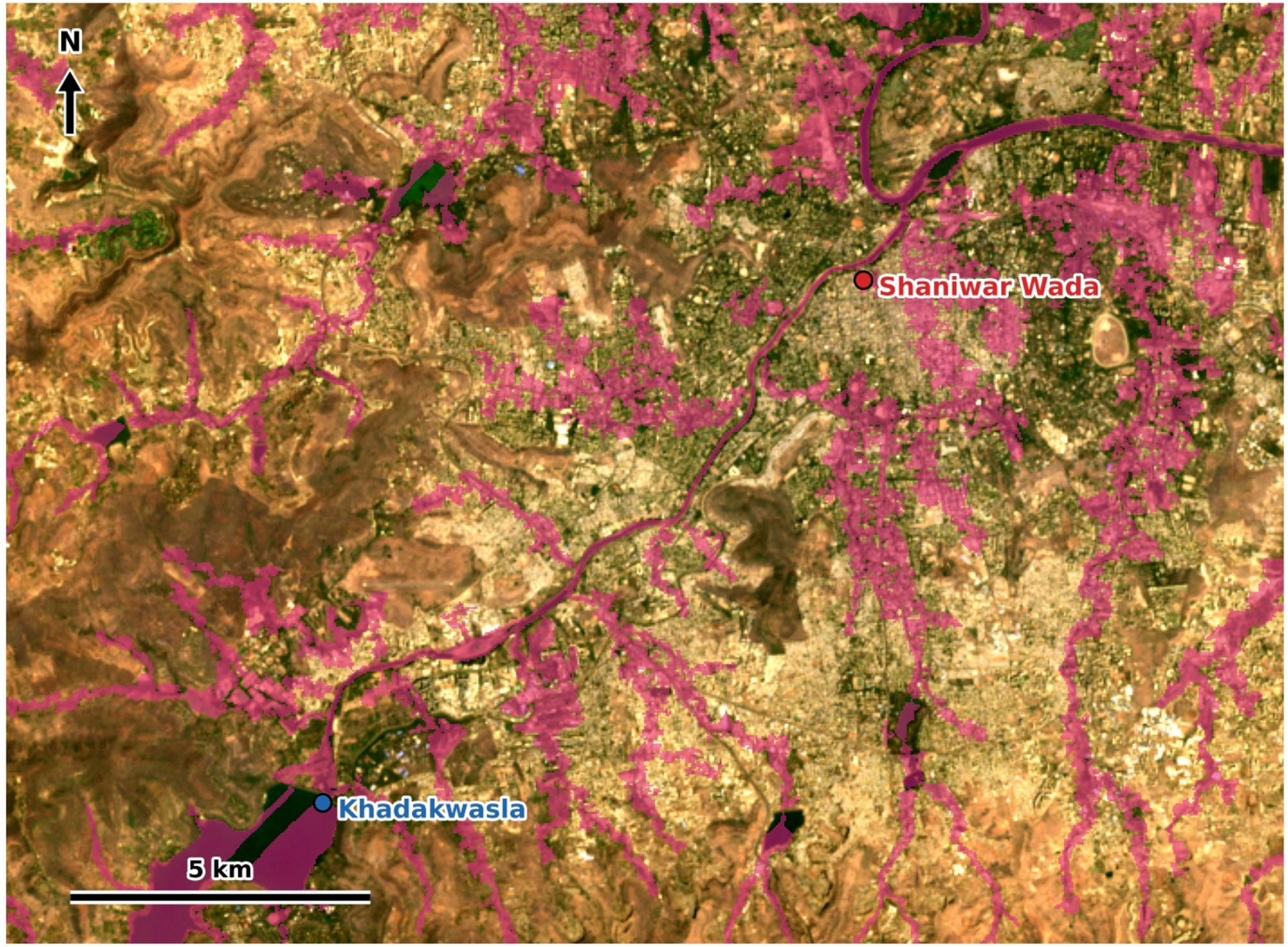
Monthly median backscatter (3 × 3 pixels around each place) from the 12 monthly Sentinel-1 products of 2020.
Water is lowest all year (it rises a little in the monsoon, when wind and rain roughen the surface). City and forest stay high and nearly flat.
The farm pixel changes most and is highest in October–December. Such differences help models where clouds hide optical images.

Terrain: elevation, slope, HAND and wetness index



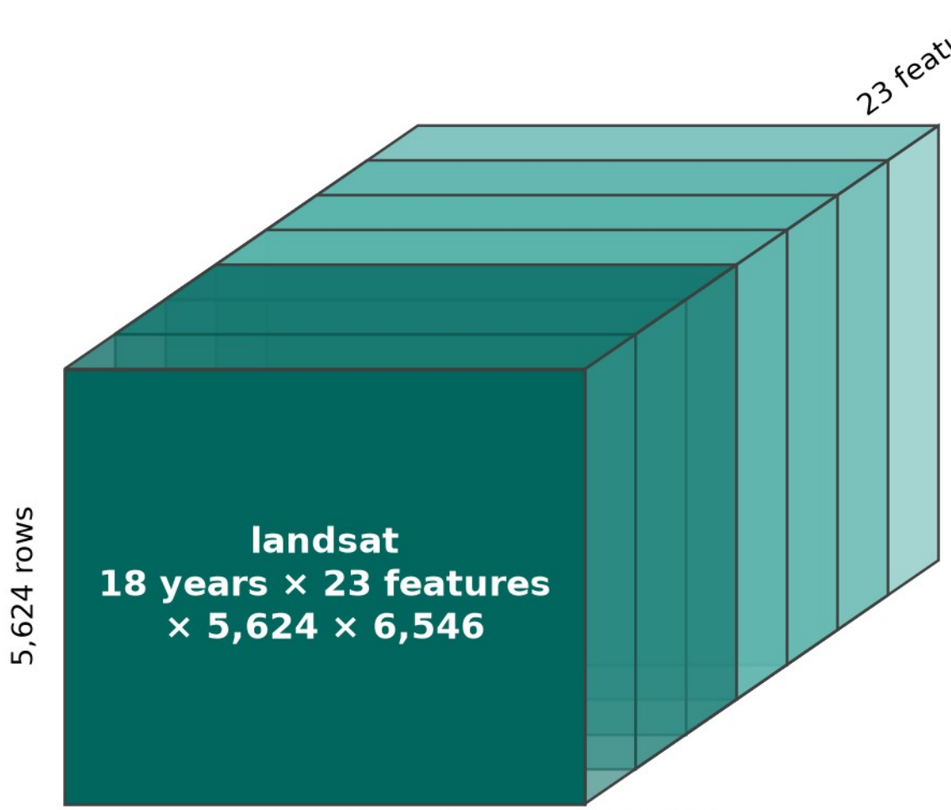
From the Copernicus 30 m surface model (2011–2015). HAND = height above the nearest drainage line (low = valley floors).
TWI = topographic wetness index (high = where water collects). HAND is biased in dense city and forest: roofs and trees raise the surface.
These layers do not change between years; the models use them to tell, for example, a valley field from a hill slope.

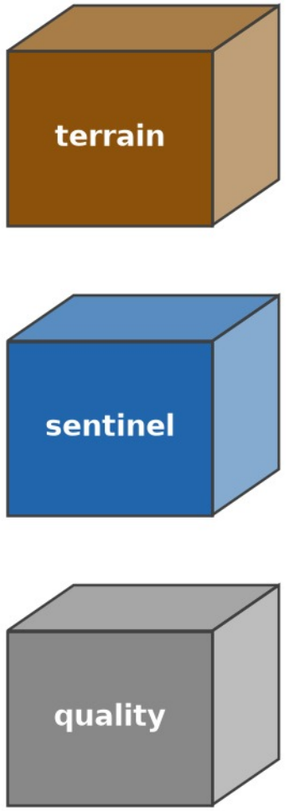
Low-lying land around the city (HAND below 5 m)



Pink = less than 5 m above the nearest drainage line (HAND < 5 m): 20.8 % of this 21 × 16 km window (2020 dry season, full 30 m).
This is a flood-exposure proxy, NOT a flood map: it shows where water would collect first, not where floods actually happened.
The 5 m cut-off is chosen for display. HAND comes from a surface model, so roofs and trees can hide low ground in the city.

The feature cube: everything a model needs, pixel by pixel





12 layers, full grid
(no year: land shape)

2020 only × 7 slots
(6 radar + 1 empty S2)

18 years × 3
(n_dry, n_post, se_dry_nir)

The 23 Landsat features per year

dry-season bands (6)

blue, green, red, nir, swir1, swir2

dry-season indices (9)

ndvi, evi, savi, mndwi, ndwi, ndmi, ndbi, bsi, nbr

post-monsoon indices (3)

ndvi, mndwi, ndmi

seasonal change (2)

d_ndvi, d_mndwi

annual (1)

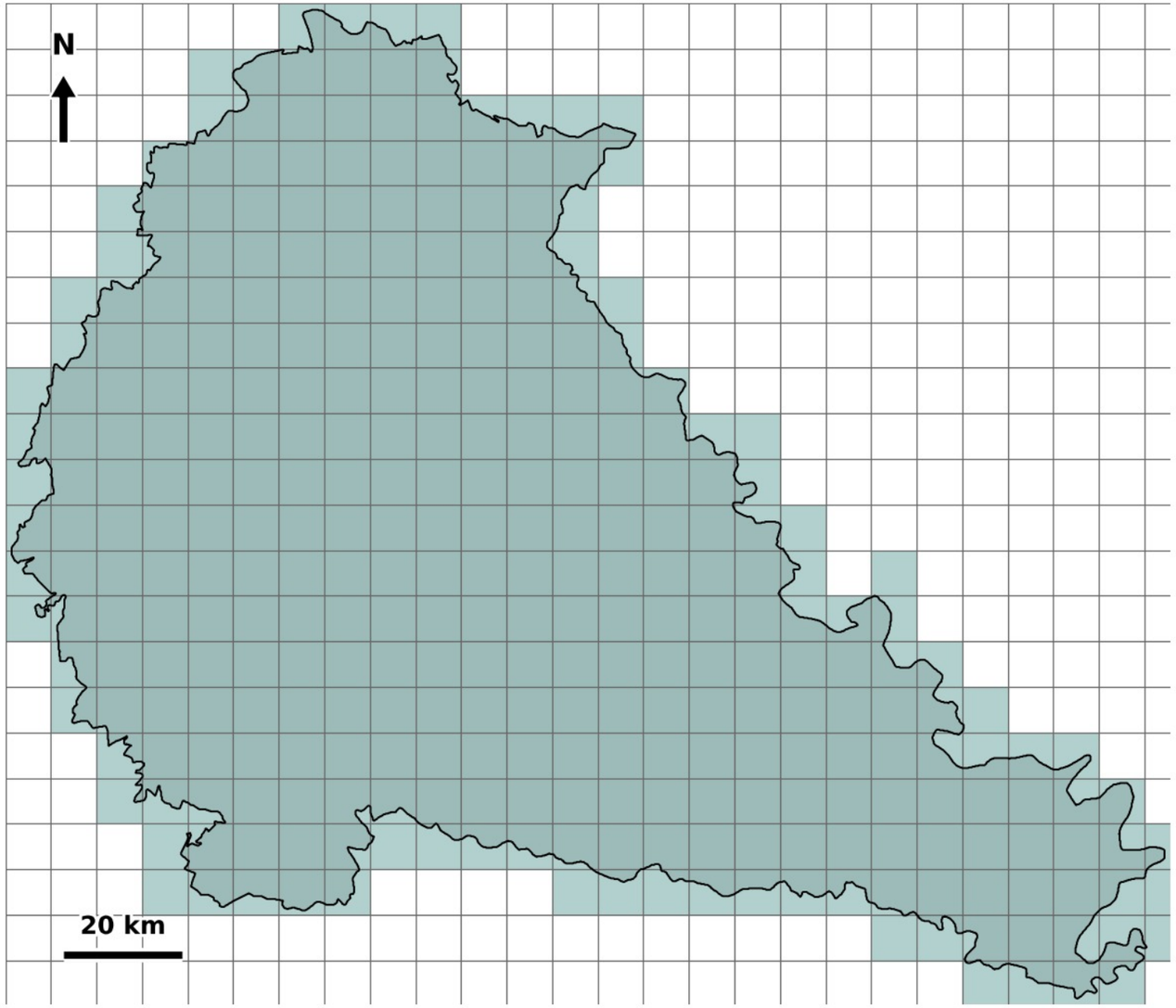
ann_ndvi

local texture (2)

sd_ndvi_5, sd_nir_5

Two versions: pune_G30.zarr (standard, 24.7 GB) and pune_G30_landsat_c2tm.zarr (Landsat 5 corrected, 24.8 GB on disk).
Every array shares the same 5,624 × 6,546 grid of 30 m cells, so one pixel's whole history can be read in one go.
The Sentinel block has 7 slots: 6 radar features plus one Sentinel-2 slot that is empty (0 % coverage; no Sentinel-2 data in the cube).

Why 17,045 files: the cube is cut into 256 × 256 tiles



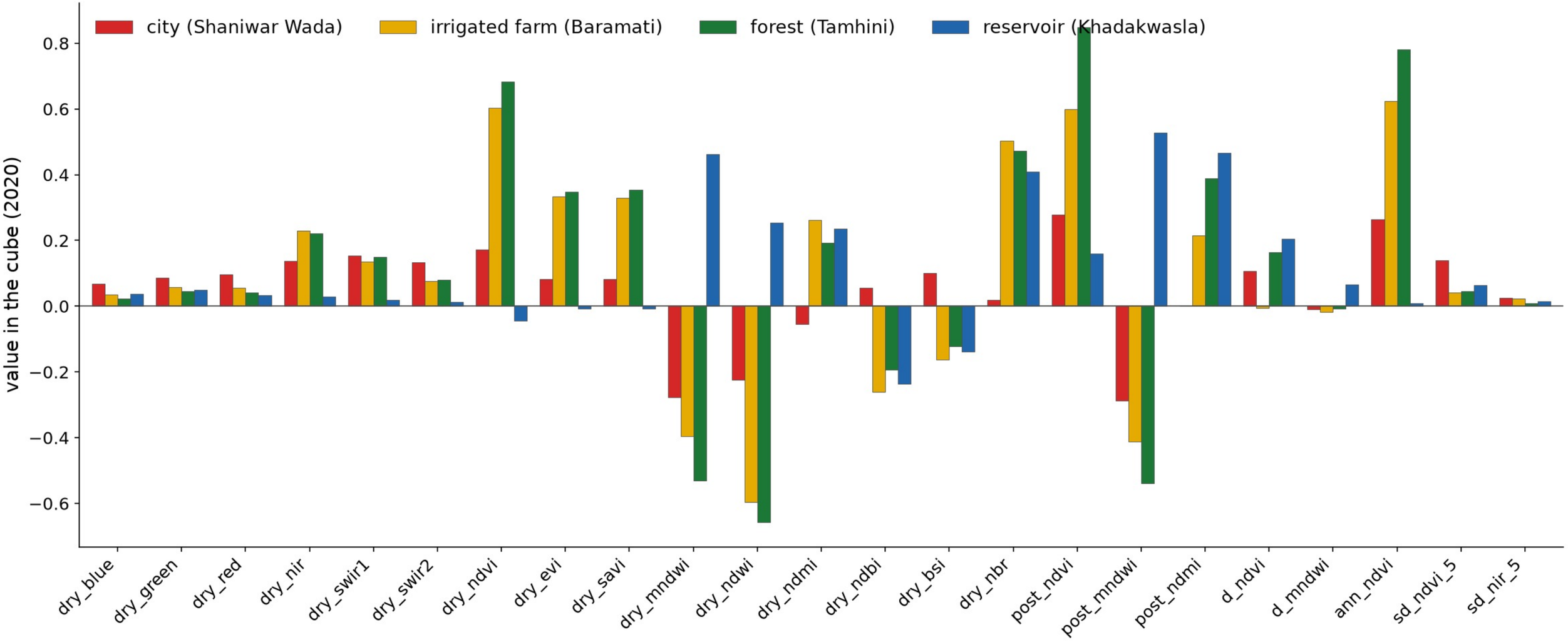
Files in pune_G30.zarr

landsat	5,833
quality	10,297
terrain	573
sentinel	325
other (metadata, coordinates)	17
total	**17,045**

Checked: the 2020 Landsat tiles on disk (324) are exactly the 324 tiles that touch the district.

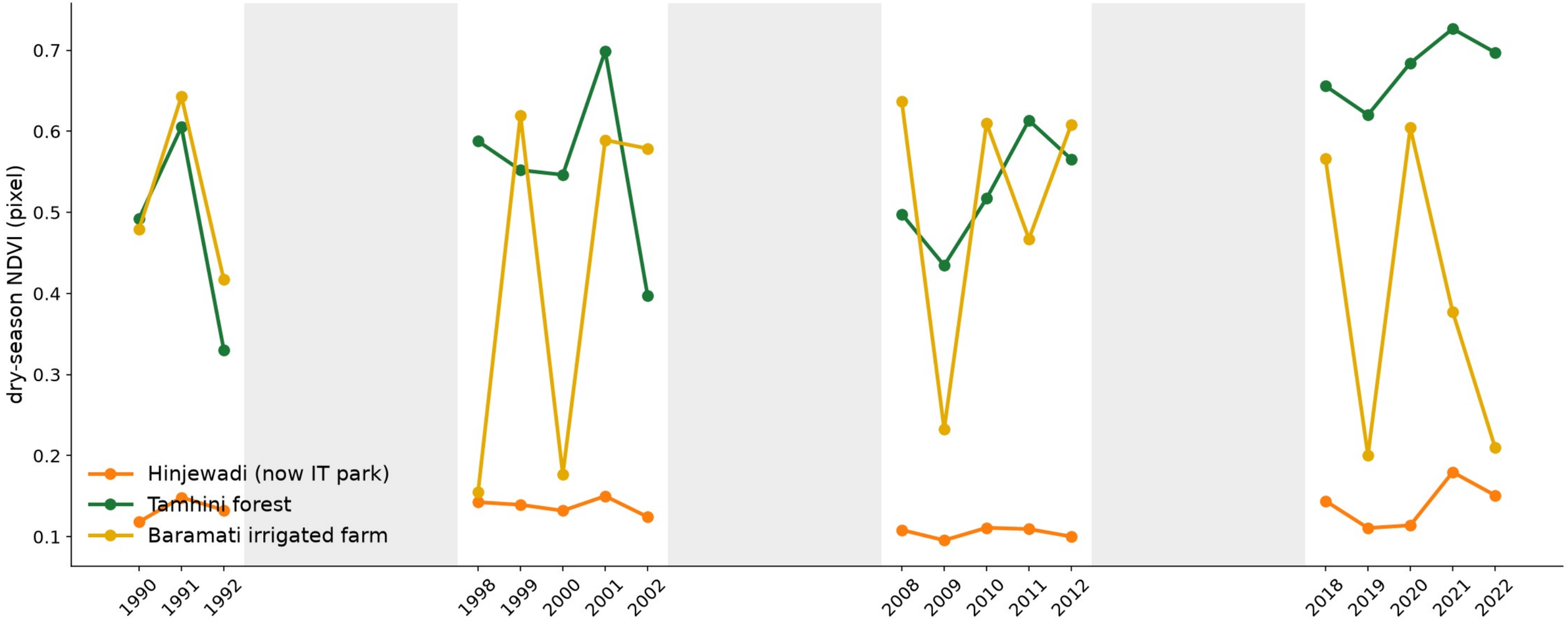
A 22 × 26 grid = 572 tiles of 256 × 256 cells (7.7 × 7.7 km) covers the image; 324 tiles touch the district (shaded). Each tile is one file. landsat: 18 years × 324 tiles = 5,832 files (empty tiles outside the district are not stored); quality: 18 × 572 = 10,296; terrain 572, sentinel 324, plus 21 metadata and coordinate files = 17,045 per cube. Two cubes → 34,090 files.

One pixel's 23 features in 2020: city, farm, forest, water



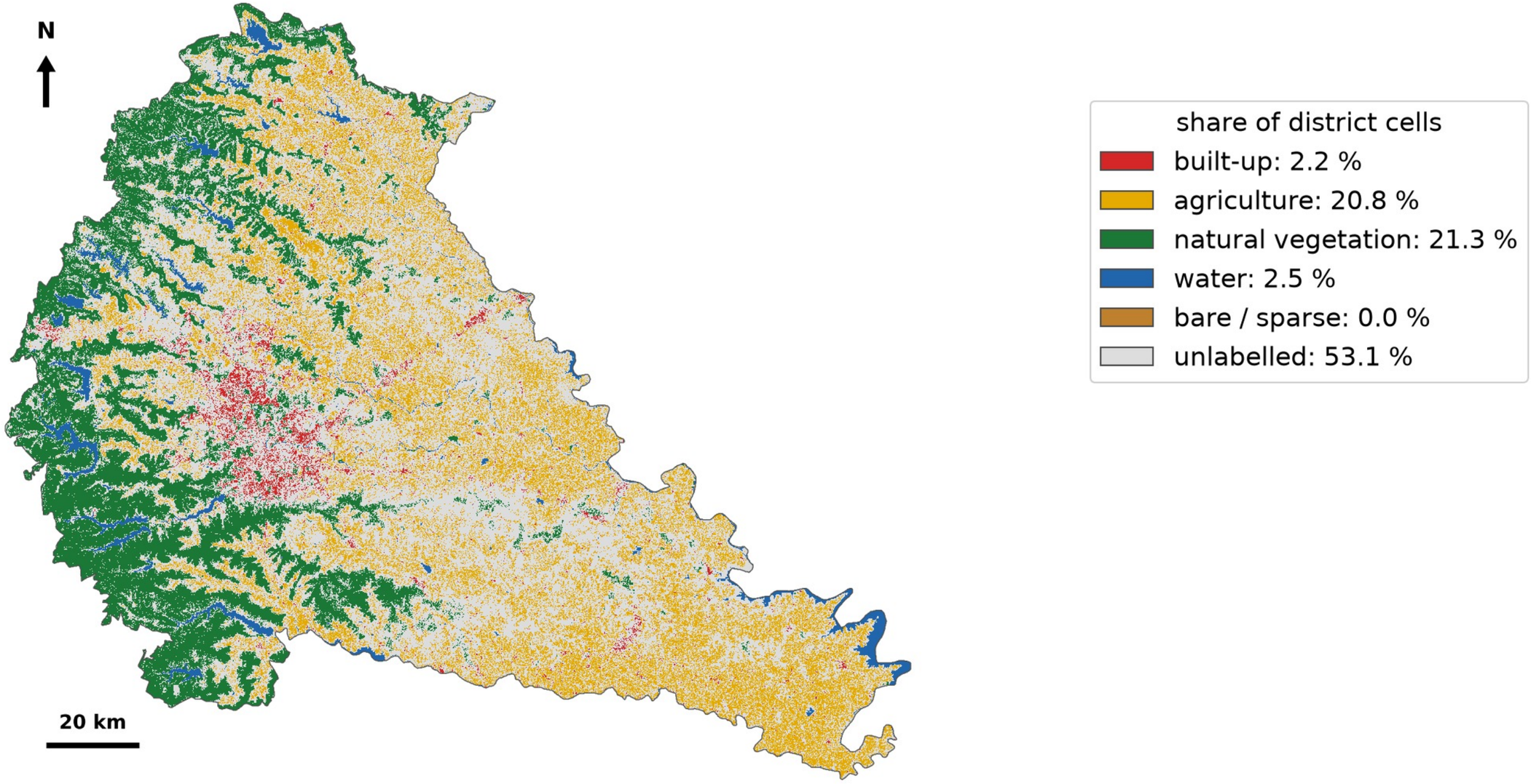
Values read straight from the feature cube (2020) at the four marked pixels. Reflectance bands are 0–1; indices run from −1 to +1.
Water: negative NDVI, high MNDWI. Forest: high NDVI and NIR, greener still after the monsoon. City: low NDVI, positive NDBI (built-up index).
The irrigated farm is green even in the dry season (dry NDVI about 0.6). Models learn from patterns like these across all 23 features.

Dry-season NDVI over all 18 years at three pixels



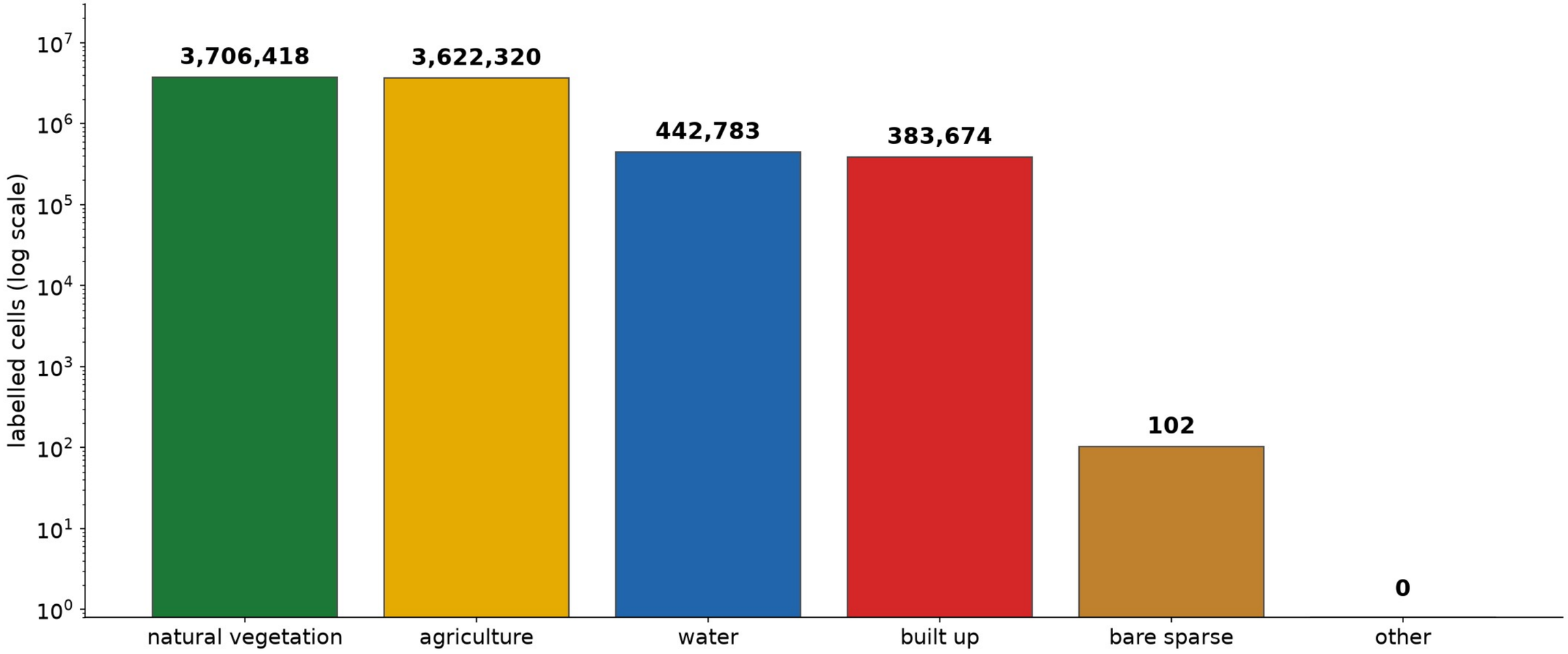
Dry-season NDVI of each pixel in every year of the cube (grey gaps = years not in the record).
Hinjewadi is low in every year (bare dry ground before, roofs after: neither is green in the dry season).
The forest is highest in most years; the farm swings a lot from year to year.
Caution: part of the change between years may come from different sensors and different rainfall years, not only from land change.

Silver labels: automatic labels, a training option, NOT a validated map



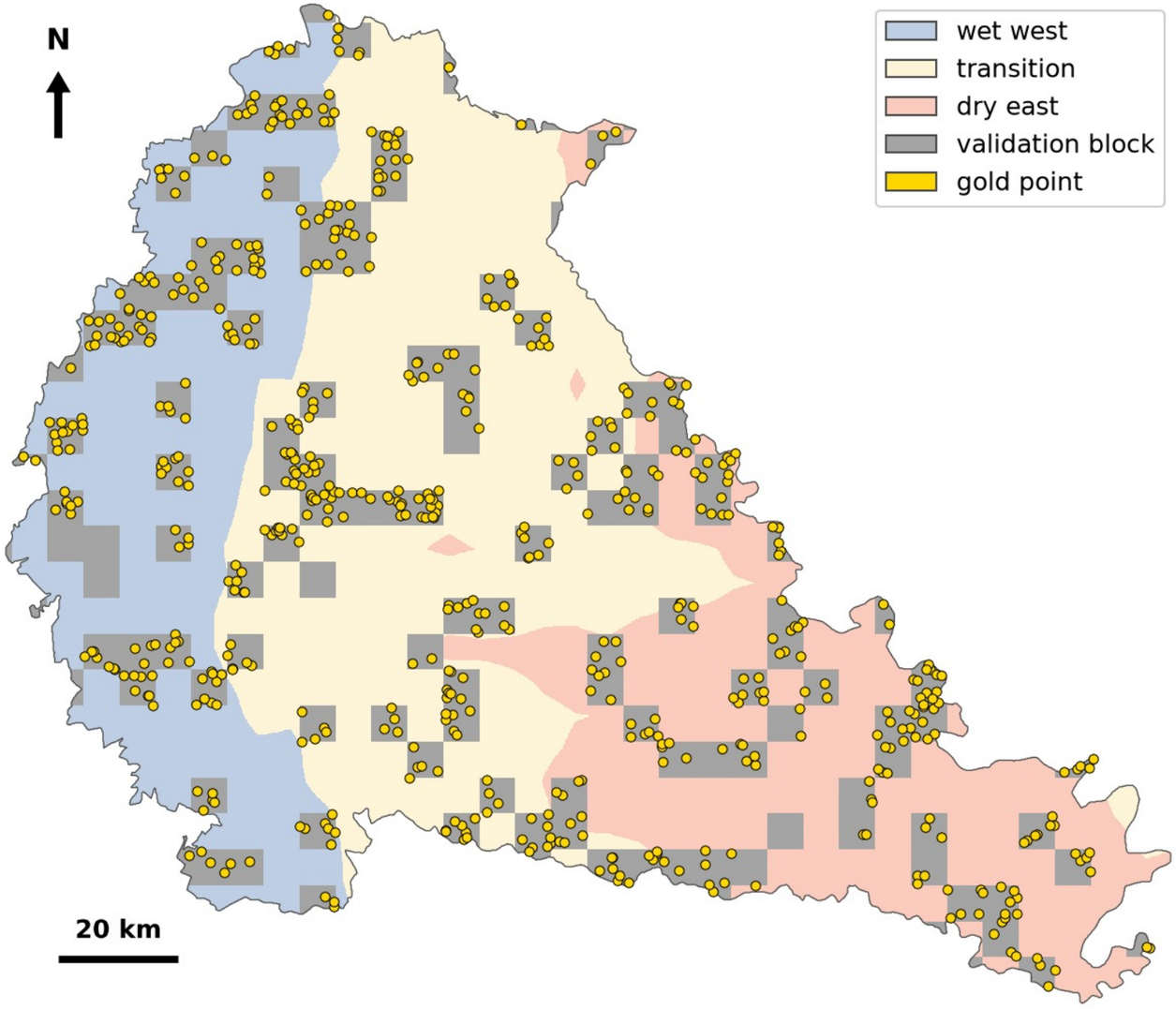
Automatic labels, a training option, NOT a validated map: a cell is labelled only where four 2020–21 products agree (ESA WorldCover 2020, 2021; Esri 2020, 2021; unanimous, purity ≥ 0.78): 46.9 % of the district (8,155,297 cells). Grey = no agreement. One candidate set of training labels (arm A of test C1); never used to measure accuracy.

Silver labels: how many cells per class

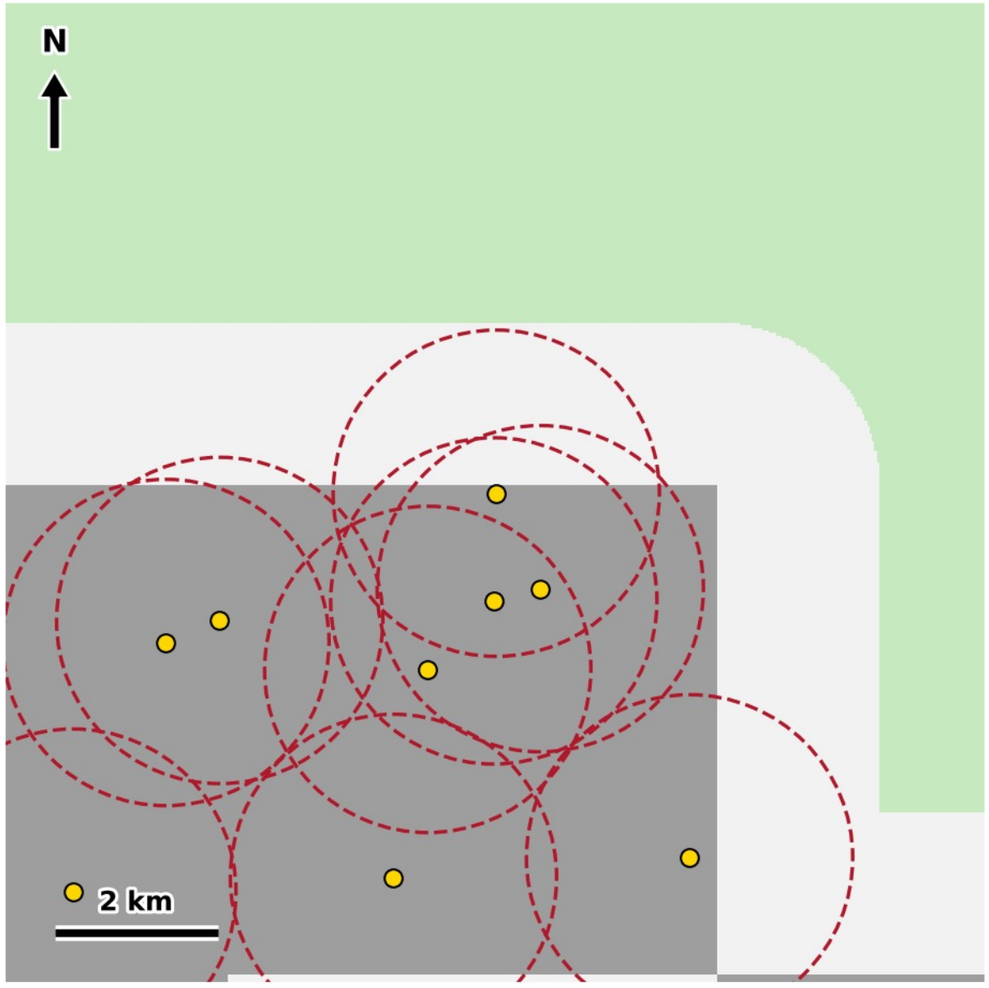


Labelled cells per class in the district silver labels (counts from the product metadata; the raster gives the same numbers).
Note: only 102 bare/sparse cells and 0 'other' cells: the four products almost never agree on bare ground.
Log scale: each grid line is ten times the one below. Automatic labels, a training option, NOT a validated map.

The gold validation design: 641 points inside validation blocks

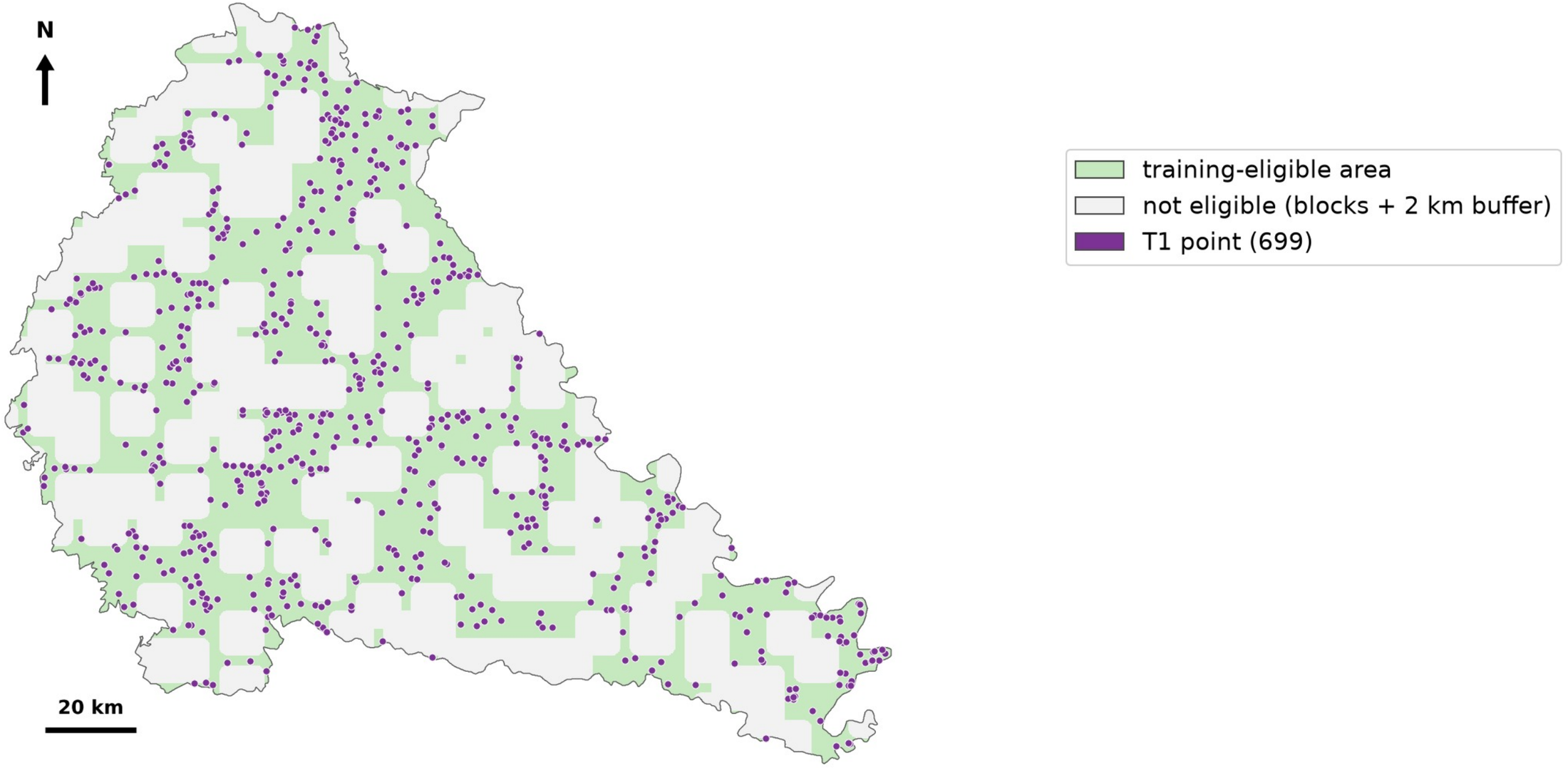


Zoom 12 × 12 km: grey = block, green = training-eligible, light = neither (buffer); red circles = 2 km around gold points



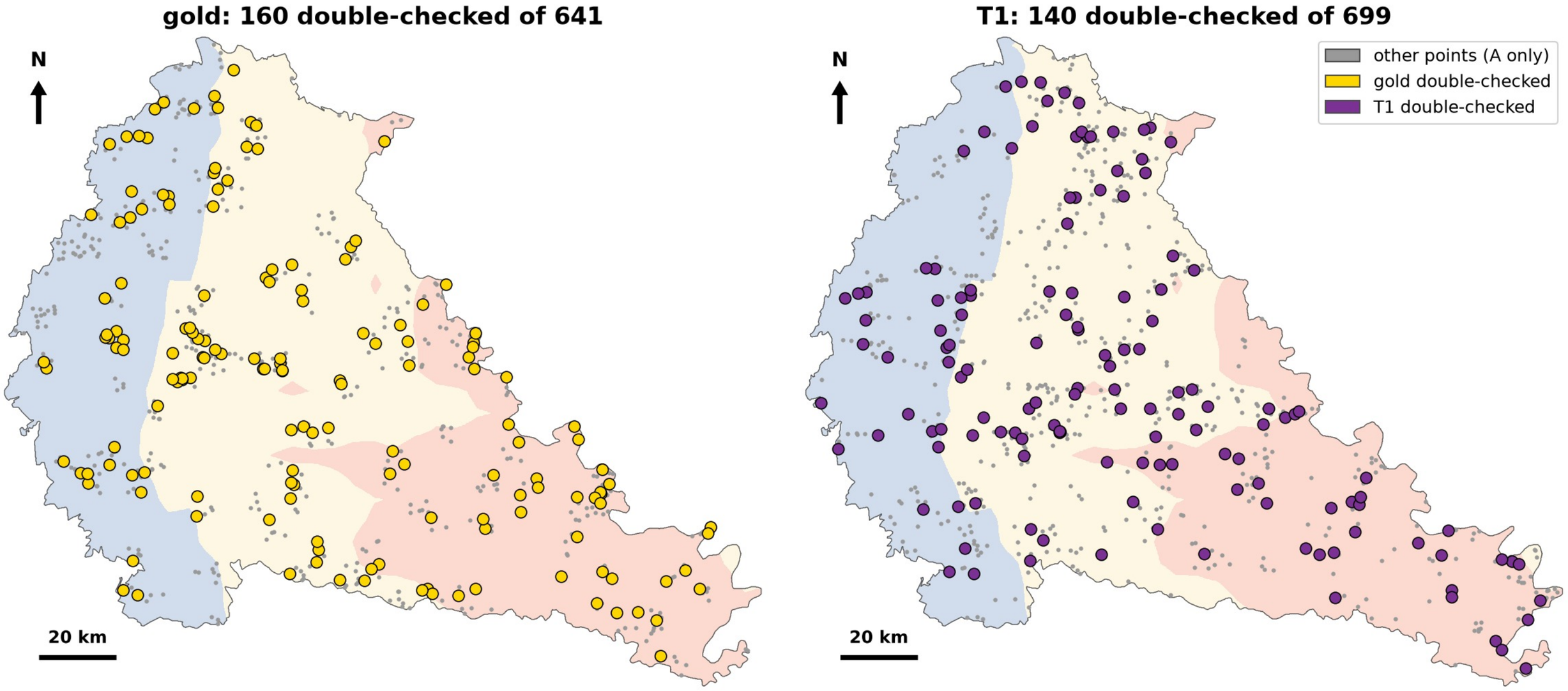
641 points (coordinates from the gold kit form) in 6 × 6 km validation blocks (grey), 26.7 % of the district; the points fall in 125 blocks.
By rainfall region: wet west 191, transition 260, dry east 190. Every point is inside a block.
2 km rule: no training cell within 2 km of a gold point (nearest: 2.01 km), so training and testing data never touch.

T1 training points: 699 points in the training-eligible area



Green = training-eligible area (46.3 % of the district): outside the validation blocks and at least 2 km from them. All 699 T1 points lie in it.
By rainfall region: wet west 166, transition 371, dry east 162. Nearest T1 point to any gold point: 2.19 km.
Coordinates from the T1 kit form. Interpreters label each point for 2020 and 2000.

Double-checked points: labelled independently by two people



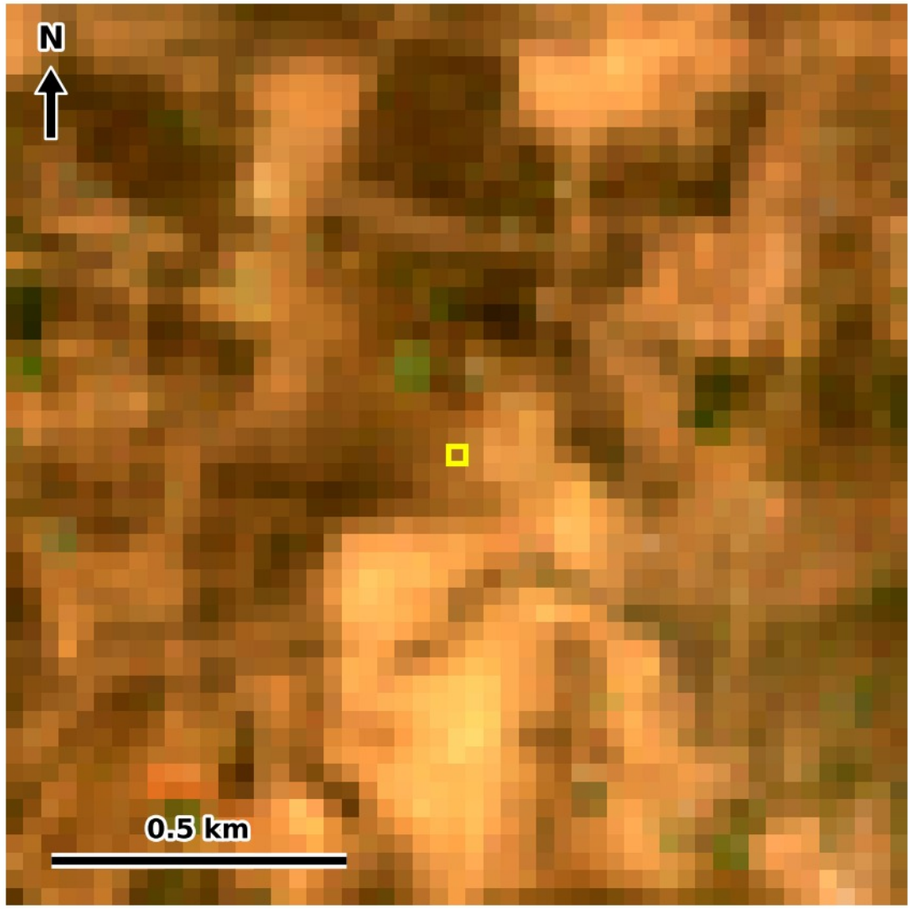
Large markers: B-kit points, labelled again by a second interpreter who never sees the first one's answers: 160 gold, 140 T1 (about 20 %).
Gold B by region: wet 38, transition 71, dry 51. T1 B: wet 31, transition 72, dry 37.
Agreement between the two (Cohen's kappa) shows how reliable the labels are. Checked: every B point is also in its A kit.

What an interpreter gets: a blind form and a square to label

anon_id	lat	lon	dominant_class	dominant_share_3bins	second_class	confidence_1to3	imagery_source	imagery_date	status	notes
P4-0001	18.391362	74.270465								
P4-0002	19.215850	73.767216								
P4-0003	18.691585	73.810318								
P4-0004	18.278134	74.722222								
P4-0005	18.575513	74.547457								

Header fields shown are the 2020 ones (e.g. dominant_class_2020).
... plus 24 more columns (the same 8 fields for 2000, 2010 and 1990), and 636 more rows.

P4-0001: 1.5 × 1.5 km, every pixel = 30 m



Left: the first 5 rows of the gold form (35 columns; only anon_id, lat and lon are filled). No class, stratum, region or prediction is shown.
Right: where the first point's 30 m square lies, on the 2020 dry-season Landsat composite at full resolution (yellow square = one cell).
Interpreters (protocol v2) use Google Earth Pro historical imagery, not this 30 m picture, which only shows where the square is.

Labelling progress: nothing labelled yet

Gold: 641 points × 4 epochs (2020, 2000, 2010, 1990)

0 / 2,564

+ 160 points double-checked · effort ~54–106 h
(27–53 h for 2020 + 2000, the same again for 2010 + 1990)

T1 training: 699 points × 2 epochs (2020, 2000)

0 / 1,398

+ 140 points double-checked · 1,678 interpretations in all · effort ~28–56 h

Quality bar (gate G2)

☐ ≥ 80 % of points labelled for 2020 and 2000 (no image = recorded, never guessed)

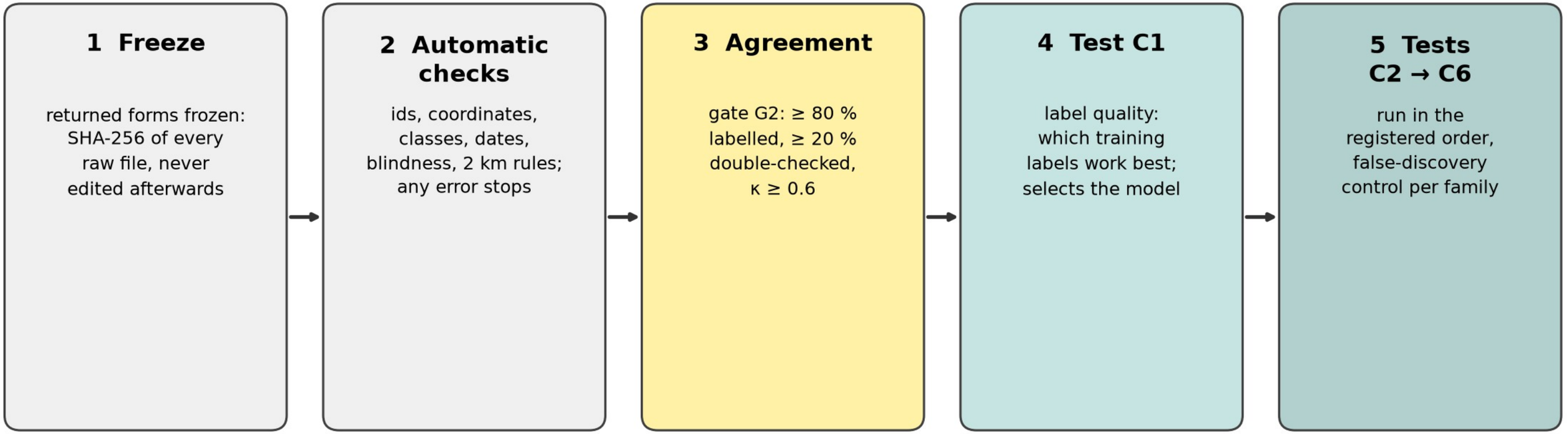
☐ ≥ 20 % double-checked

☐ agreement: Cohen's κ ≥ 0.6

☐ every disagreement adjudicated and logged

No label set has been frozen yet (no manifest in results/phase7/manifests/gold or /T1); all six tests are registered, not yet run.
Effort figures are the planning estimates in the protocol documents (1–2 minutes per interpretation), not measurements.
Plan under protocol v2 (human interpreters). Decision D-R3 (8 Oct): AI-proposed, researcher-verified labels instead; v3 drafted, not adopted.

After labelling: the fixed order of work



Each step must pass before the next starts; the runner refuses to start a test whose prerequisites fail.
C1 picks the model used by C2–C6 using the TUNING part of the T1 labels only, never the gold results.
Every run is recorded once; a re-run would be a new, separately registered version.

The six registered questions (C1–C6): all not yet run

test	question in plain words	what is compared	labels needed	status
C1	Do better training labels give more accurate historical maps?	5 label sets: automatic (1 or several years), validated, T1 human, T1 + filtered automatic	gold (4 epochs) + T1	**not yet run**
C2	Does correcting Landsat 5 to the newer sensor improve 1990 and 2000 maps?	c2_tm_pif_v1 correction vs no correction	gold 1990, 2000	**not yet run**
C3	Do era-aware models beat one model for all years?	era-specific vs one unified model	gold (4 epochs)	**not yet run**
C4	Does the model work in a region it never saw?	leave-one-rainfall-region-out (target: macro-F1 ≥ 0.70, gap ≤ 0.10)	gold 2020	**not yet run**
C5	Are the model's confidence levels trustworthy?	calibration and coverage per region and era (target: ECE ≤ 0.05, coverage ≥ 0.85)	gold 2020, 2000	**not yet run**
C6	Does adding radar and terrain help?	optical + radar + terrain vs optical only (2020)	gold 2020	**not yet run**

From the registry (experiments/registry4, P4-C1…C6@v2, status PENDING). Targets are thresholds written before any data; they are not results. 'Gold' = the validation points (slide 38); 'T1' = the training points (slide 39). C2–C6 use the model that C1 selects on the T1 tuning split.

Where we stand

Done

✓ 134 composite products for 18 years
(72 Landsat, 44 corrected Landsat, 16 Sentinel-1, 2 benchmark)

✓ terrain (12 layers) and two feature cubes

✓ 94.3 GB on disk in 34,931 files

✓ 34,965 SHA-256 fingerprints (95.3 GB fingerprinted)

✓ 217 tests passed (2 skipped) in the last full run

✓ C1–C6 registered and pre-run checks wired

Remaining

☐ human labels: gold 641 points × 4 epochs,
T1 699 points × 2 epochs

☐ freeze, automatic checks, agreement (G2)

☐ tests C1, then C2–C6

☐ validated maps with error bars
(accuracy with confidence intervals)

From product metadata, an on-disk count of the three data folders, the SHA-256 list (results/phase7/mac/handoff/)
and the last full test run (commit b970de2, 8 Oct 2026). The fingerprint list also covers 34 quarantined and external files.
Plan under protocol v2 (human interpreters). Decision D-R3 (8 Oct): AI-proposed, researcher-verified labels instead; v3 drafted, not adopted.